

Polygenic scores

Konstantinos Hatzikotoulas

Agenda

1. Human genetics recap
2. GWAS recap
3. Complex traits
4. Polygenic scores

1

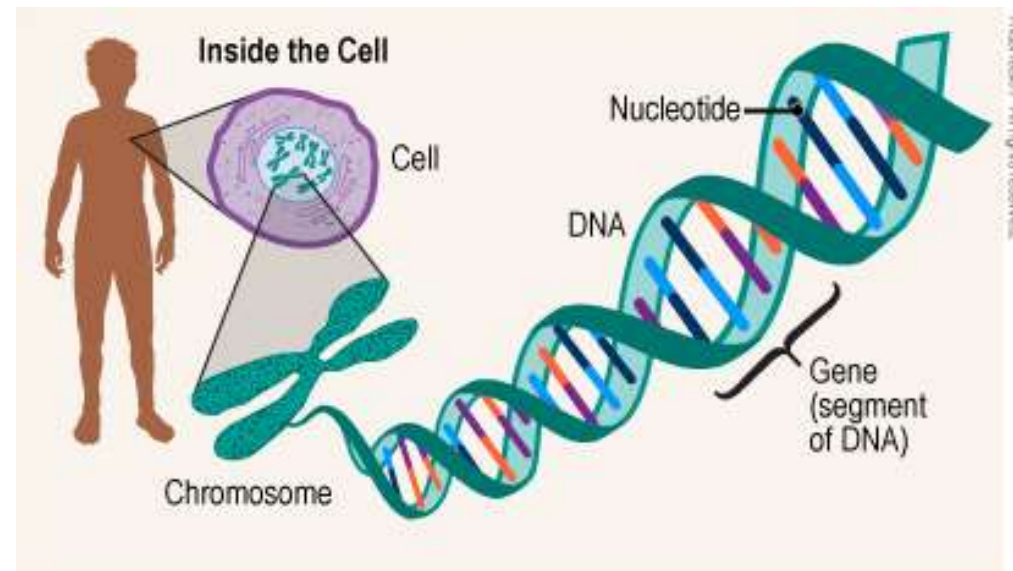
Human genetics recap



➔ What is a gene?

A gene is a **sequence of nucleotides** in DNA or RNA that **encodes the synthesis of a gene product**, either RNA or protein.

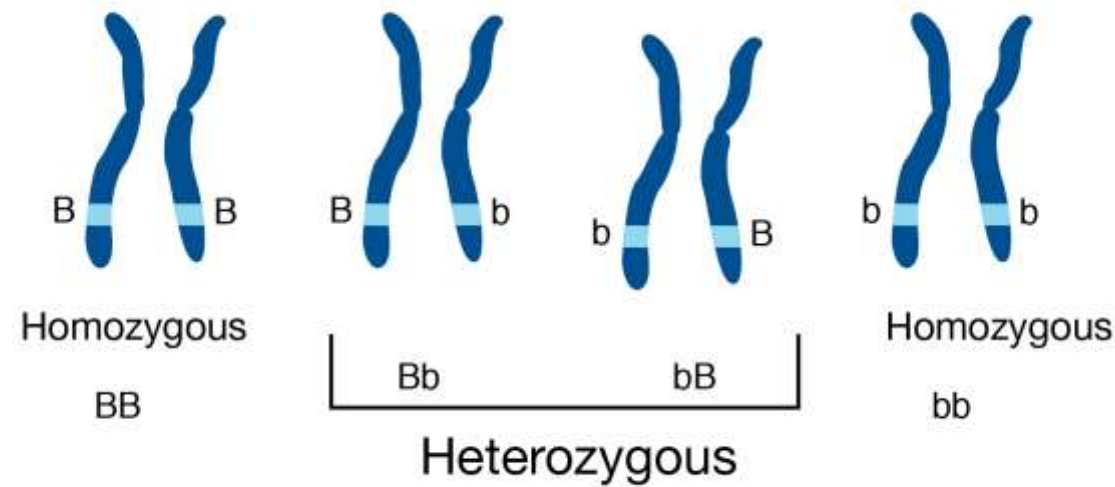
A genome region that includes all of the sequence elements necessary to **encode a functional transcript** and specifies a trait.



H → What is an **allele**?

Allele: different forms of the same gene that determines an organism’s phenotype. It is represented by letters.

Humans are **diploid organisms**, which means that they have **two alleles at each genetic position**, or locus, with one allele inherited from each parent.



Allele b count			
BB	bB	Bb	bb
0	1	1	2

⇒ What is a **genotype?
What is a **phenotype**?**

Genotype vs Phenotype

GENOTYPE

The genotype is an organism's genetic information.

BB

homozygous dominant

Bb

heterozygous

bb

homozygous recessive

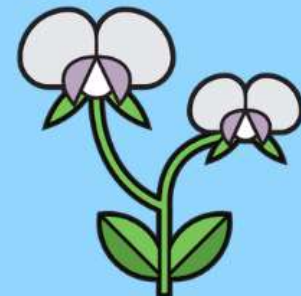
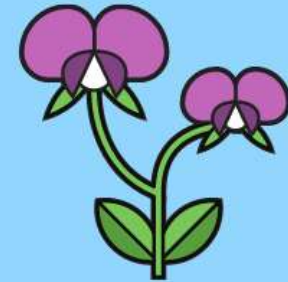
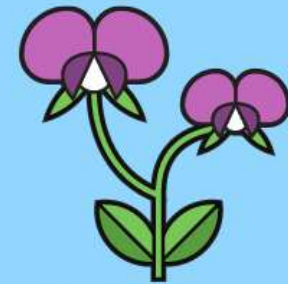
PHENOTYPE

The phenotype is the set of observable physical traits.

purple

purple

white



EPISTASIS

An interaction between the products of two genes in which one of the genes modify the phenotypic expression produced by the other.

Offspring phenotypic ratio



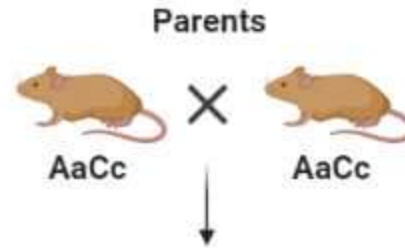
Brown
9/16



Black
3/16



Albino
4/16



Offspring genotypes

	AC	aC	Ac	ac
AC	 AACC	 AaCC	 AACc	 AaCc
aC	 AaCC	 aaCC	 AaCc	 aaCc
Ac	 AACc	 AaCc	 AAcc	 Aacc
ac	 AaCc	 aaCc	 Aacc	 aacc

Epistasis

Coat color in mice

A= brown (dominant)

a= black (recessive)

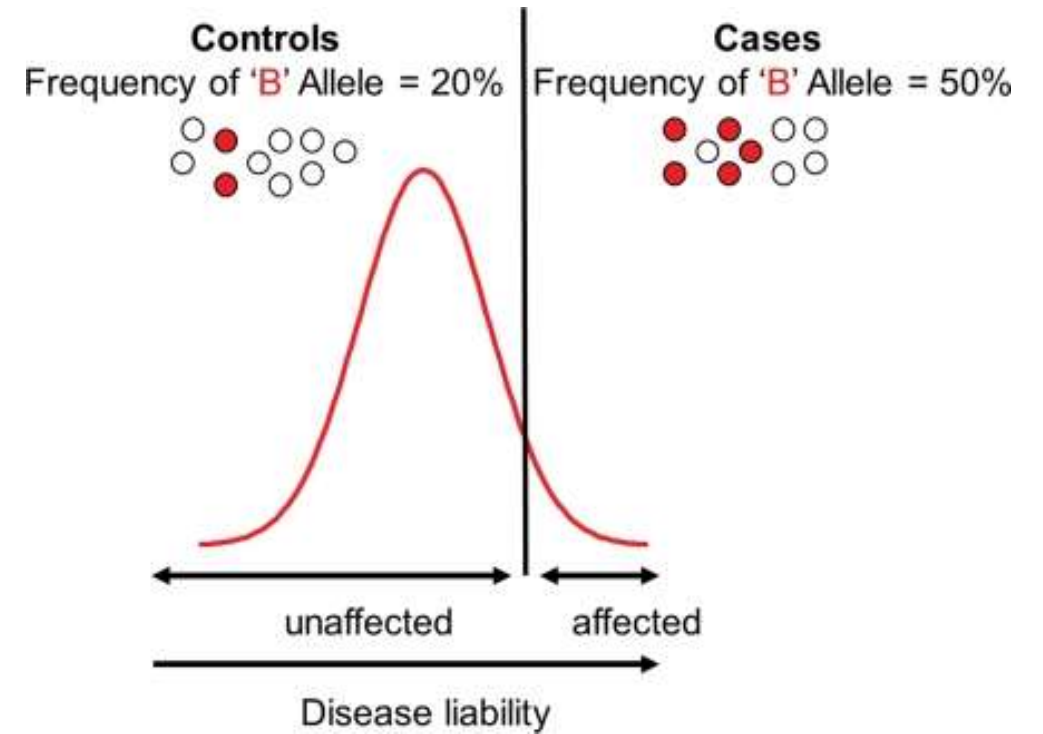
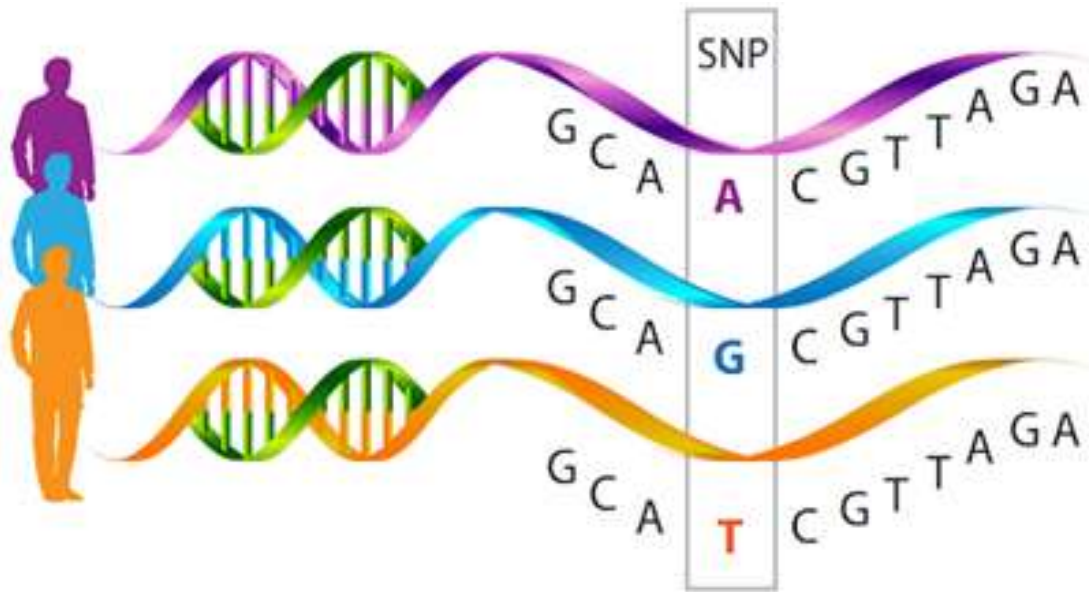
C= pigment (dominant)

c= no pigment (recessive)

➔ What is a **SNP**?

What is a **risk/effect allele**?

Single-nucleotide polymorphism



2

Genetic association studies



Modelling

- Is there an association between the **phenotype** (disease, continuous trait) and the **genotype** ?

$$\textit{phenotype} \sim \beta \times \textit{genotype} + \epsilon$$

$$\begin{bmatrix} \textit{pheno}_0 \\ \vdots \\ \textit{pheno}_n \end{bmatrix} \quad \begin{bmatrix} A/T \\ \vdots \\ T/T \end{bmatrix} = \begin{bmatrix} 1 \\ \vdots \\ 2 \end{bmatrix}$$

$= \{0,1\}$ (case-control)
 $\in \mathbb{R}$ (quantitative) $\sim \mathcal{N}(0,1)$

$= \{0,1,2\}$ (genotype, directly typed)
 $\in [0,2]$ (dosage, imputed)

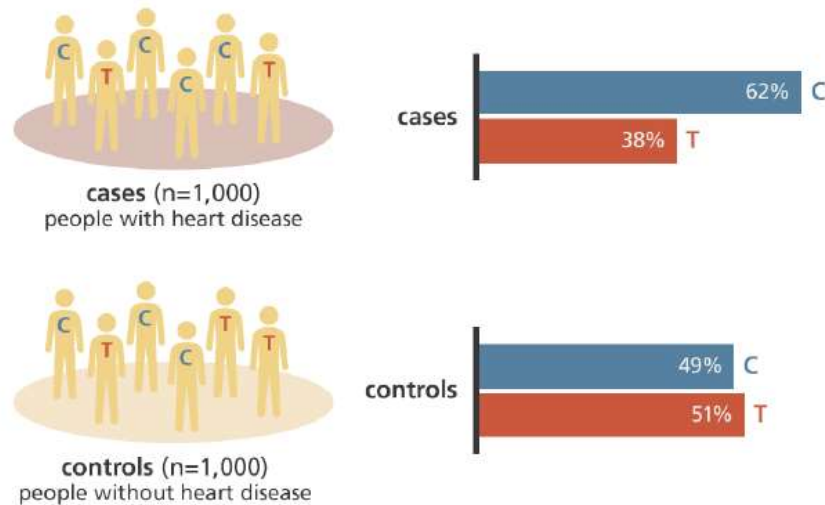
$$\begin{bmatrix} 0.965 \\ \vdots \\ 1.816 \end{bmatrix}$$

- For each variant, association test $\rightarrow$ if $p \leq 5 \cdot 10^{-8}$: variant significantly associated
- Estimation of the effect of the variants: β or Odds Ratio (OR)

Case/control studies

Odds ratio (OR): *how much more likely are you to be a case if you carry the risk allele ?*

➤ Per genotype g and disease Y , we compute the odds $O = \frac{p}{1-p} = \frac{p_{Y=1|g}}{1-p_{Y=1|g}} = \frac{p_{Y=1|g}}{p_{Y=0|g}}$



	Cases	Controls
T	380	510
C	620	490

$$O_T = \frac{380/n_T}{510/n_T} \quad O_C = \frac{620/n_C}{490/n_C}$$

$$OR = \frac{n_{\text{affected carriers}} \times n_{\text{healthy non-carriers}}}{n_{\text{healthy carriers}} \times n_{\text{affected non-carriers}}}$$

$$OR_{C/T} = \frac{620 \times 510}{490 \times 380} = 1.7$$

Case/control studies

OR = ratio of the odds of the two alleles

- OR>1: the allele is 'deleterious'
- OR<1: the allele is 'protective'

Statistical test: is the OR significantly different from 1?

- Earlier: Fisher's exact test or Chi-squared test
- Nowadays + for imputed data: linear regression or GLM

The General Linear Model (GLM)

Uses one or more regressors (independent variables) to predict an outcome measure (dependent variable)

$$Y = \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$$

Y = Dependent variable

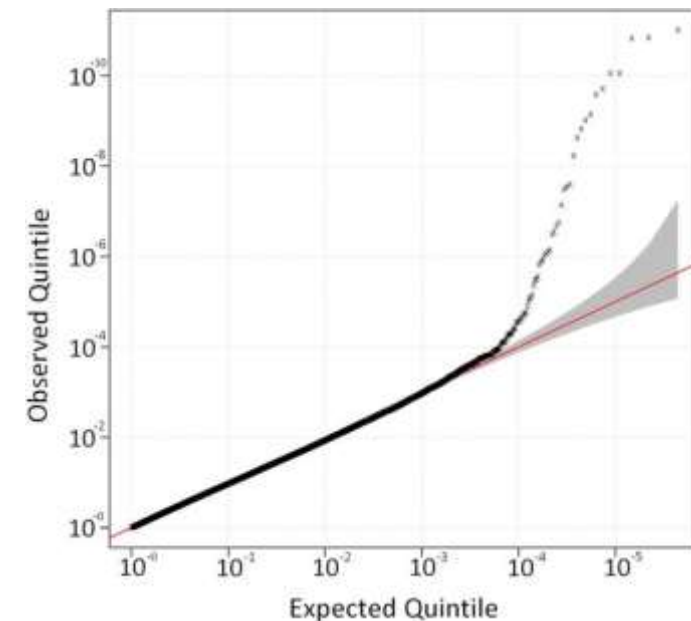
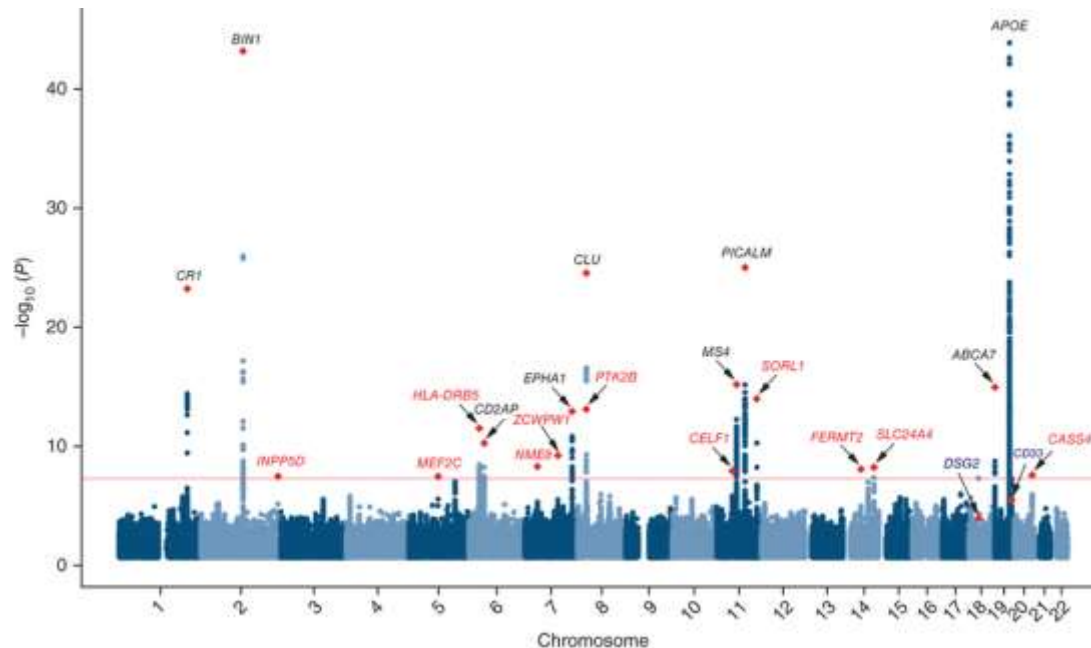
β = Beta Weights (parameter estimates)

X = Regressor

ε = Residual

GWAS results

1. Quality control (QC) of the data
2. Run model
3. Correct p-value for multiple testing (significance threshold for genomics = 5×10^{-8})
4. Visualize results (Manhattan plot)
5. Run sensitivity analysis



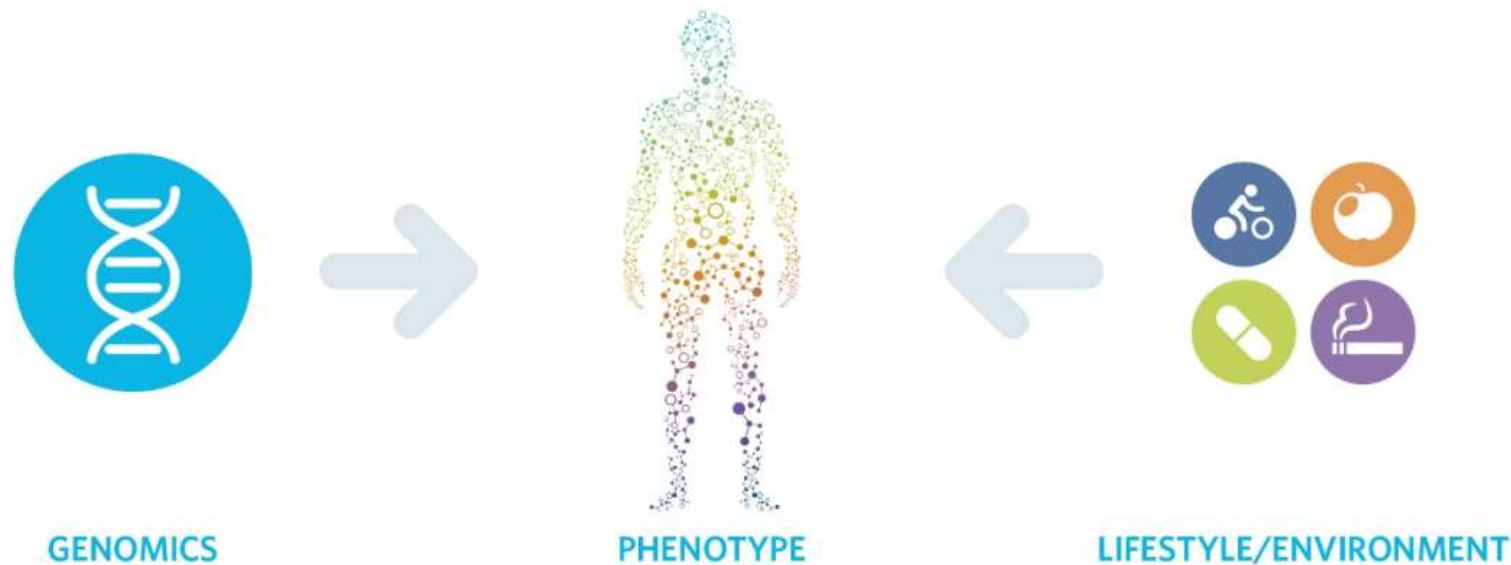
3

Complex traits



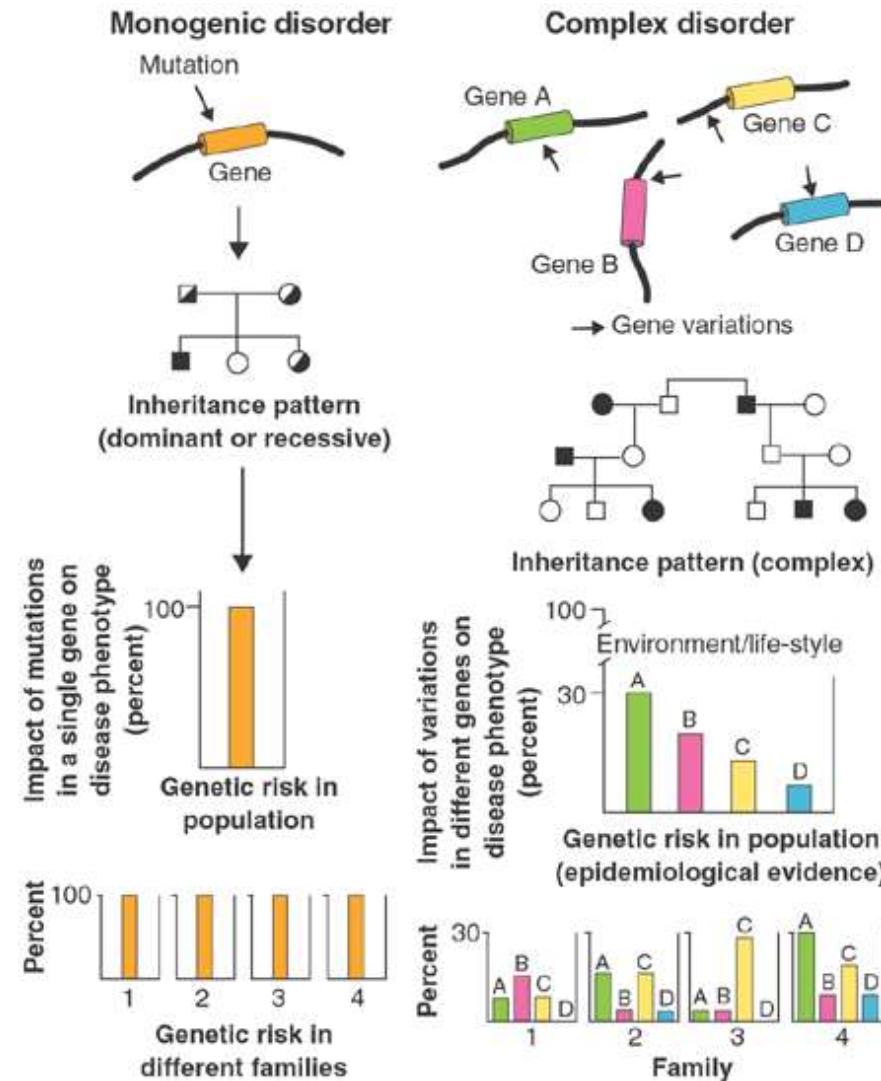
Complex traits

- Complex traits = interaction between (often many) **genetic** and **environmental** factors



- Examples: body shape, type 2 diabetes, Alzheimer's disease...
- Complex diseases tend to be common
→ Tool of choice = GWAS

Monogenic disorder vs complex traits



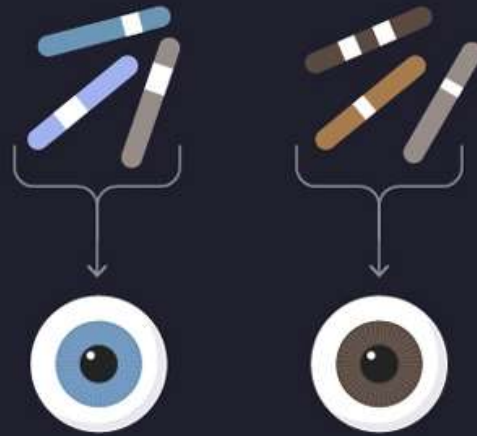
How Many Genes Are at Work?

Simple traits may be controlled by just one gene (monogenic). More complex traits are usually considered polygenic, but a new theory suggests that a better description might be omnigenic because all of the genes are involved.



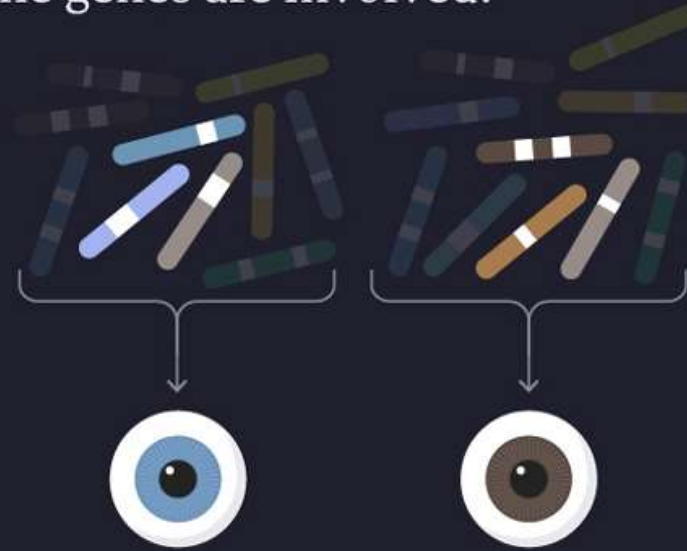
Monogenic

A single gene gives rise to a trait.



Polygenic

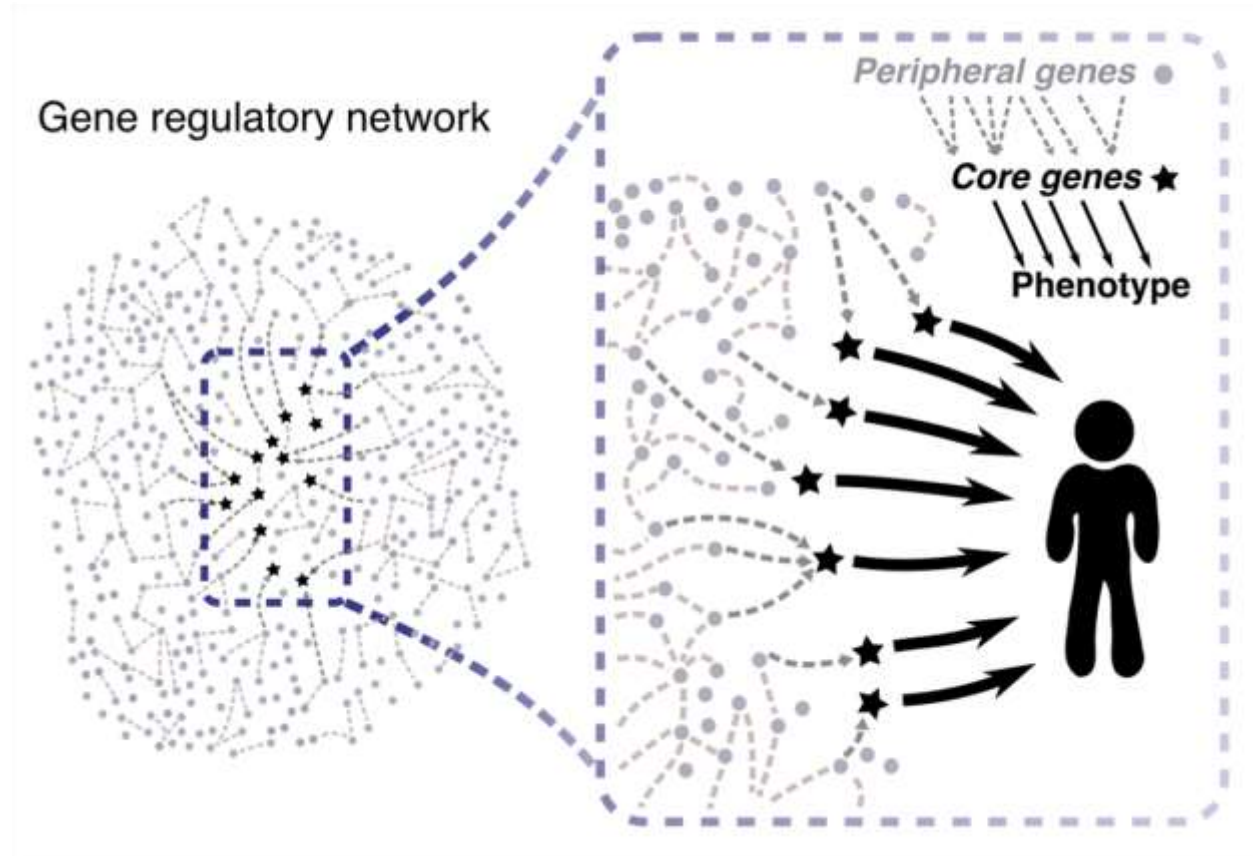
A handful of genes jointly give rise to a trait.



Omnigenic

A few core genes are essential but all the genes are involved.

Omnigenic vs Polygenic model



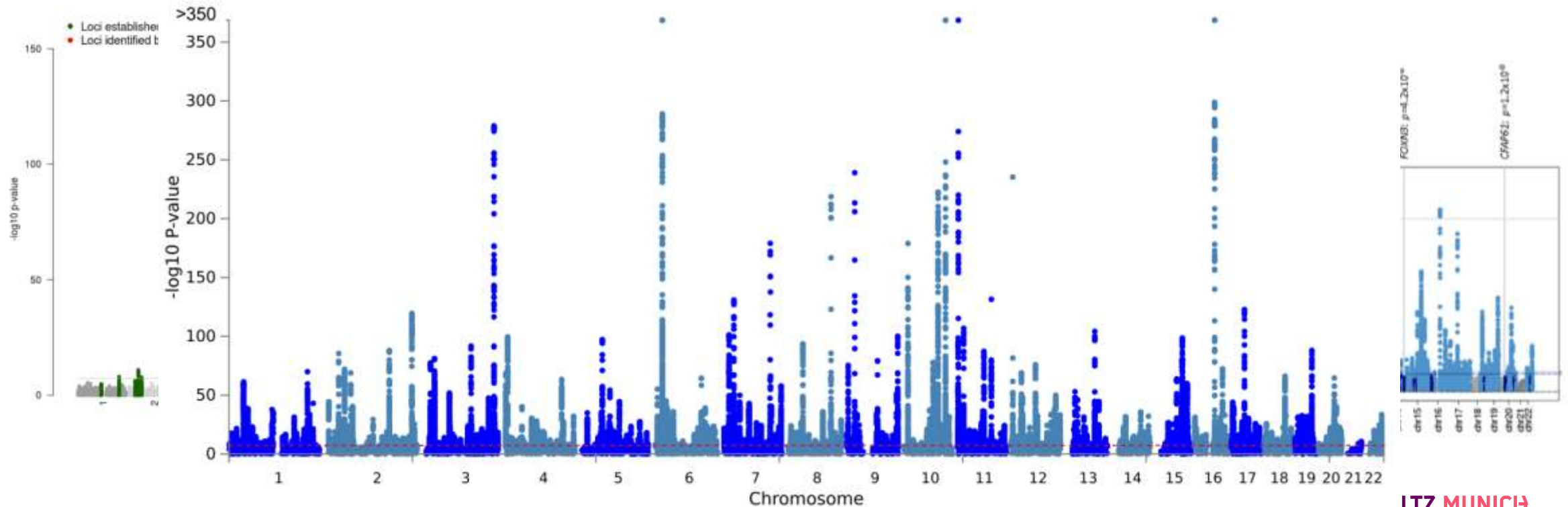
GWAS

Current tool of choice to study complex traits

Type 2 diabetes GWAS *Suzuki, Hatzikotoulas, Southam et al. Nat. 2024*

Number of cases = 428,452

Number of controls = 2,107,149



3

Polygenic scores

3.1

Introduction



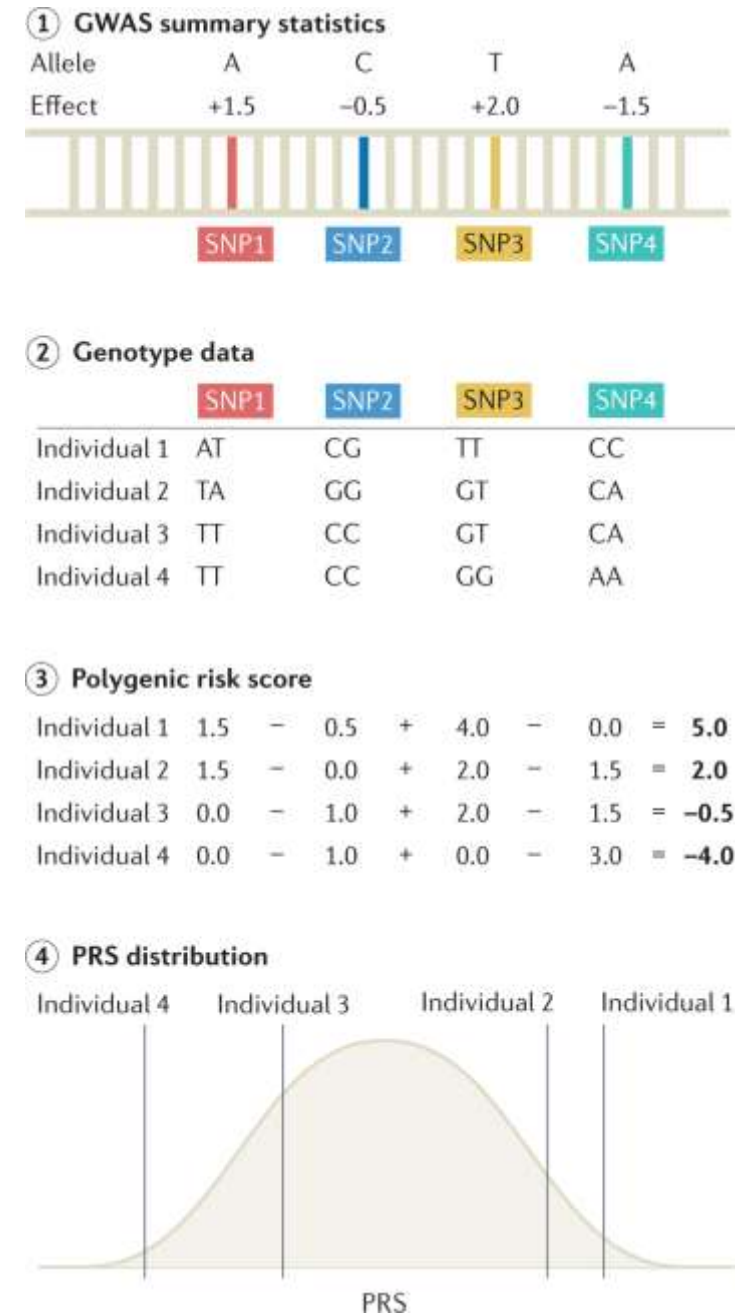
Polygenic scores

Natural follow up for complex traits:

- Influenced by many genetic variants
- GWAS → effect for each variant
- Additive model → risk increases with each copy of the effect allele

→ combine SNPs effects into a score

Used to: predict quantitative traits (= polygenic score)
or disease risk (= polygenic risk score)



Polygenic scores (PGS)

Polygenic score for individual i

Total number of SNPs included in PGS

$$PGS_i = \sum_{j=1}^{N_{snps}} G_{ij} * \beta_j$$

Effect of variant j on trait
• Estimated in GWAS

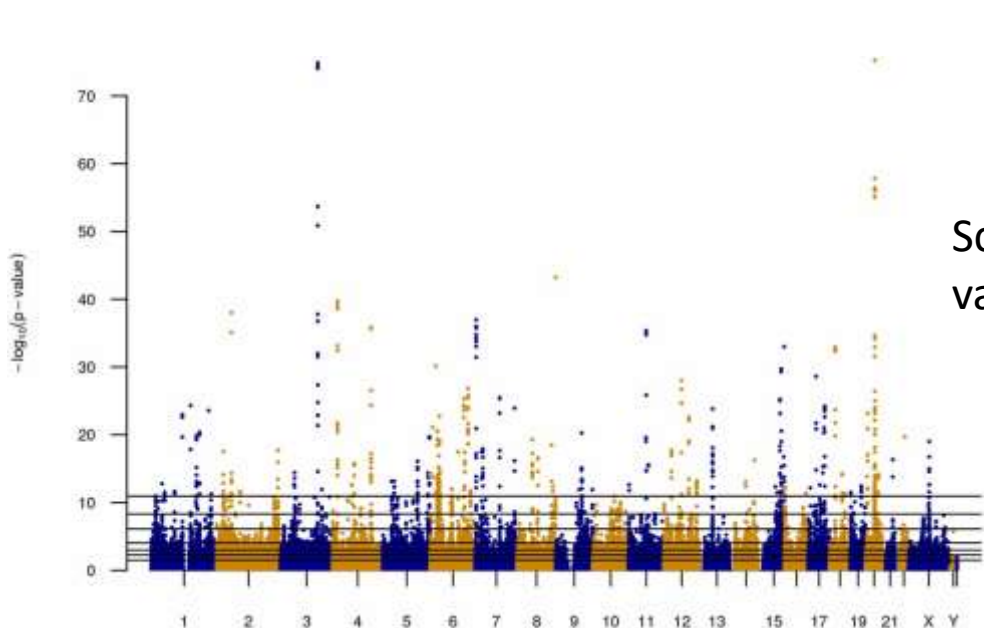
Genotype at SNP j for individual i
• Coded as 0, 1, 2 depending on the number of risk allele
• Additive model

$$PRS = \beta_1 SNP_1 + \beta_2 SNP_2 + \dots + \beta_n SNP_n$$

Effect size Number of risk alleles Number of SNPs

→ Sum of the number of risk alleles **weighted** by its effects

Polygenic scores: what kind of data do we need?



Large GWAS

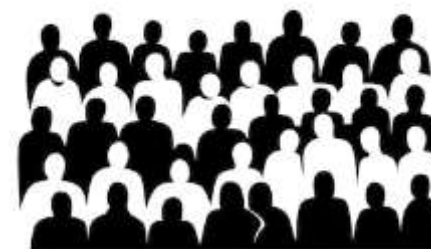
Summary-level data
= BASE

Scores of
variants



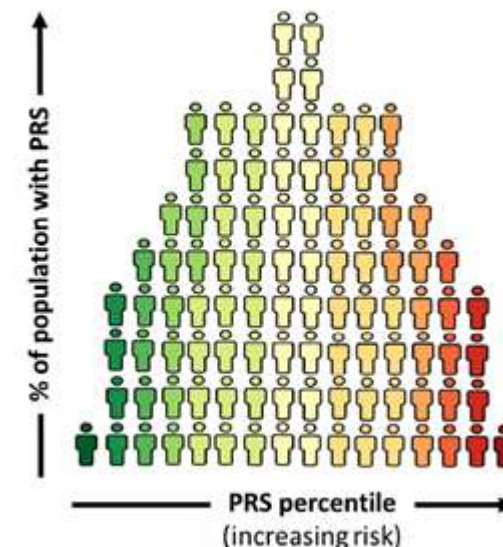
Genetic
data

$$PHS_x = \sum_i^n x_i \beta_i$$



External cohort

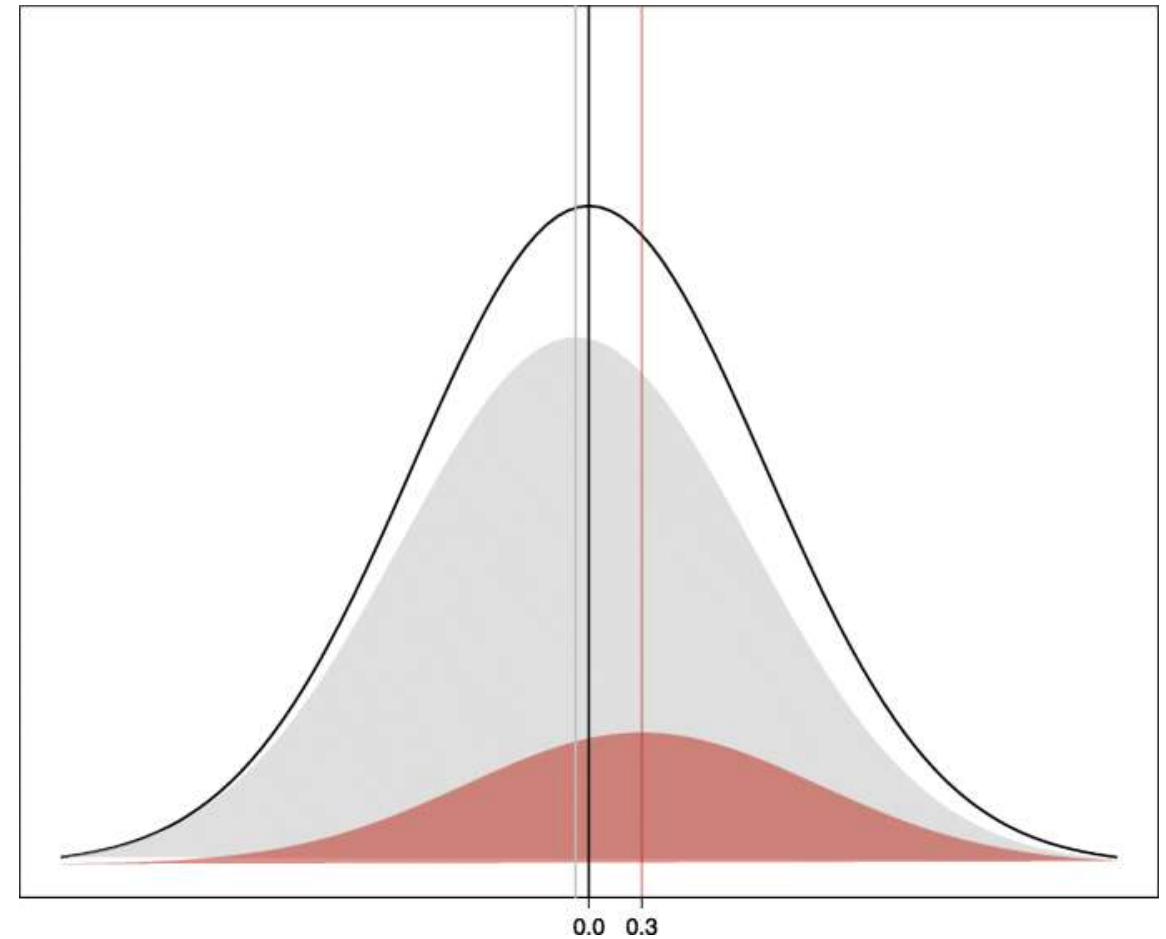
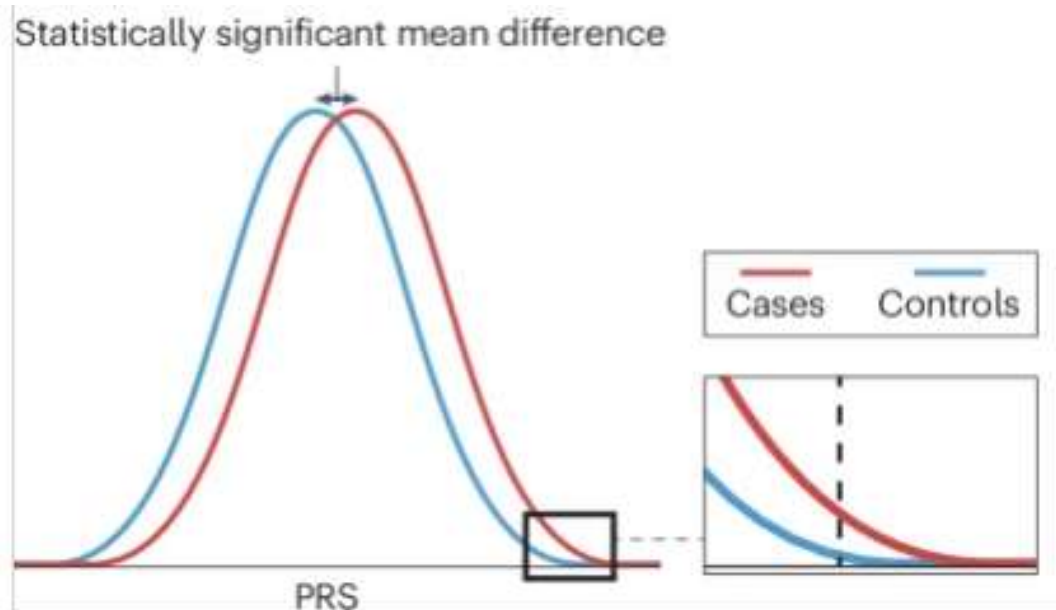
Individual-level data
= TARGET



Polygenic risk score

Case/control study

- Grey = controls
- Red = cases
- Overall mean = 0 (standardized)
- Mean difference between cases and controls directly proportional to the performance of the PGS



Kullo, I.J. Clinical use of polygenic risk scores: current status, barriers and future directions. *Nat Rev Genet* (2025).
<https://doi.org/10.1038/s41576-025-00900-8>

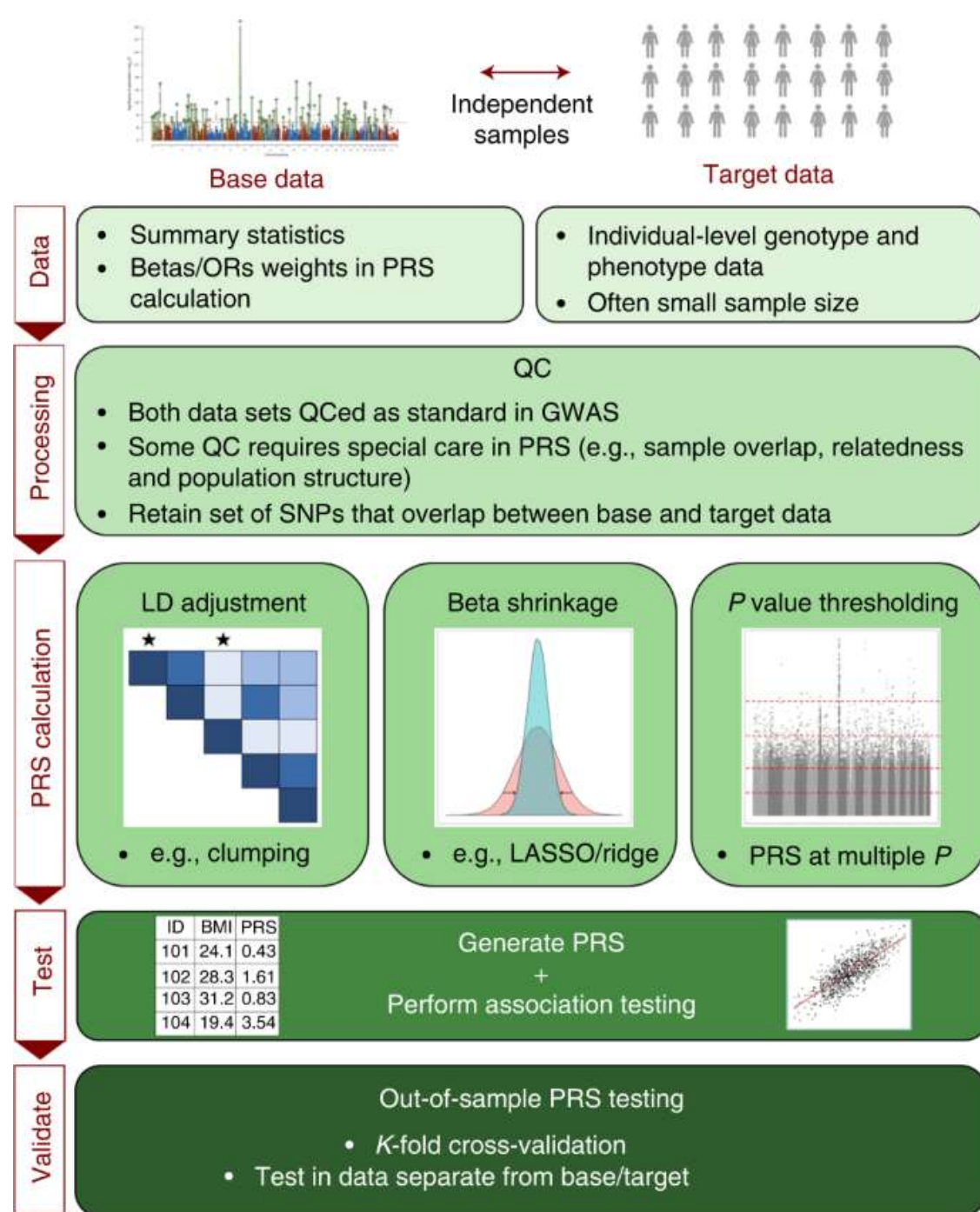
3

Polygenic scores

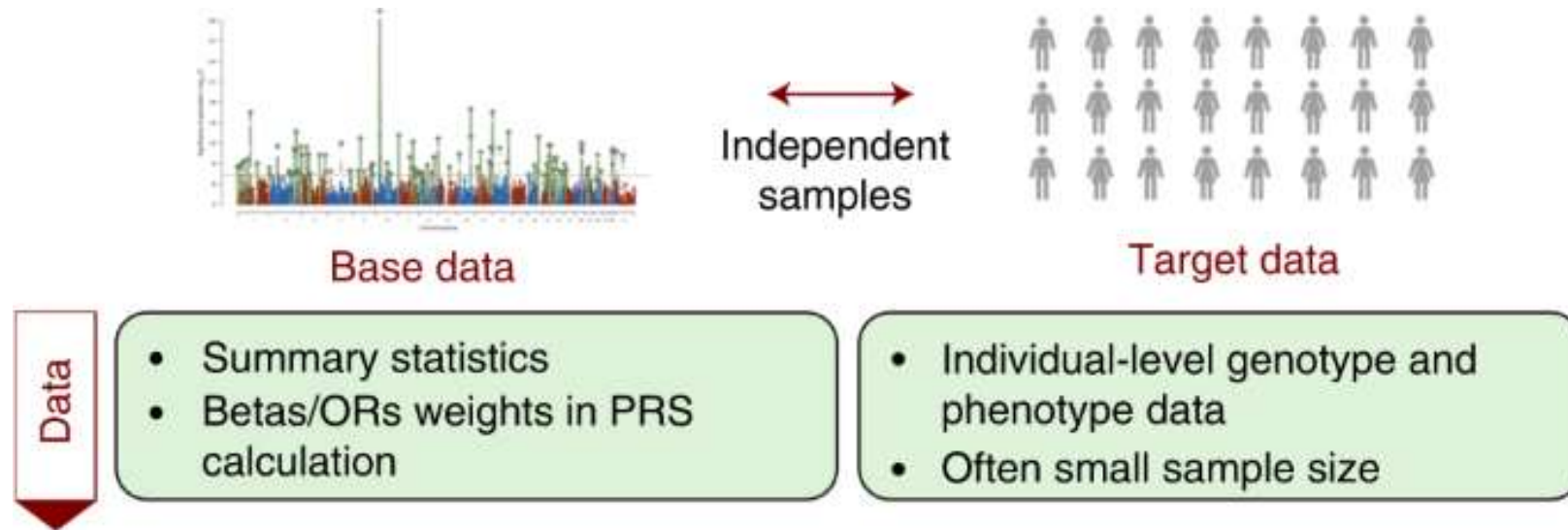
3.2

Construction





Input data



Sample used to estimate parameters for the PGS

→ Largest GWAS summary statistics

→ We need:

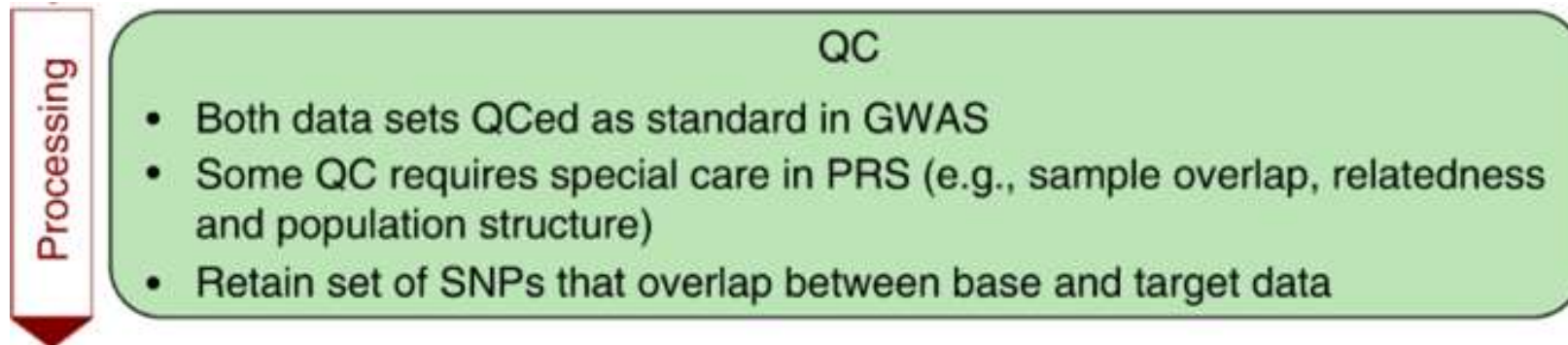
- Effect sizes of the variants: betas/OR
- standard errors
- p-values

Sample where we will apply the PGS

→ Individual level data (genotype data)

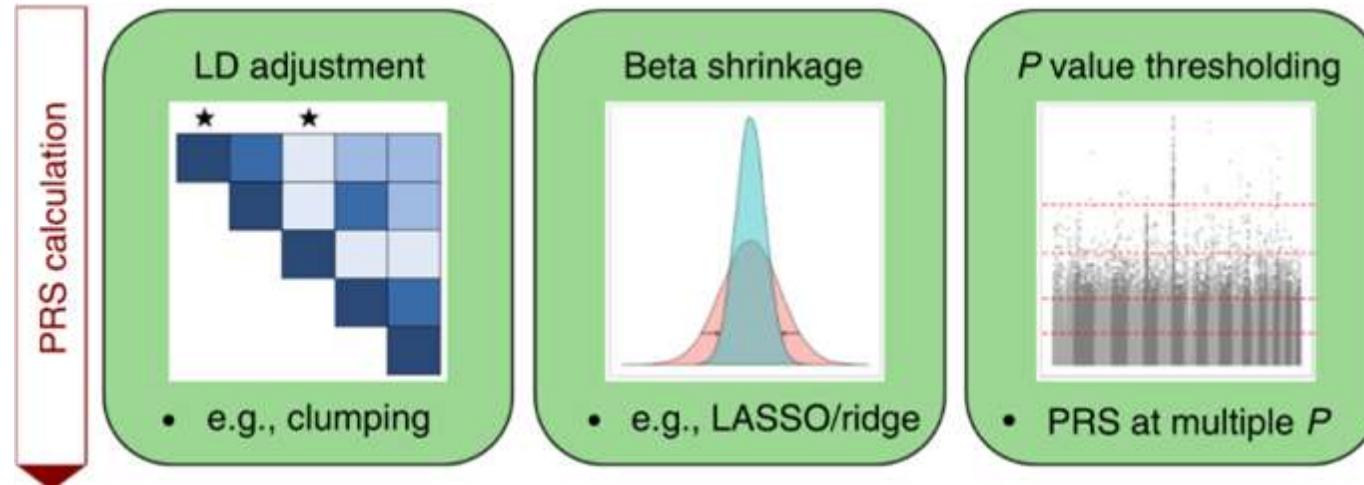
→ Often smaller sample size compared to Base data.

Data processing



- **No sample overlap** between base and target data
→ Could lead to inflation of effects: 'overfitting'
- Need **homogeneity** between base and target samples
→ Hypothesis = samples underlie the same genetic architecture and environmental conditions
→ E.g. same population structure/ancestry

PGS calculation



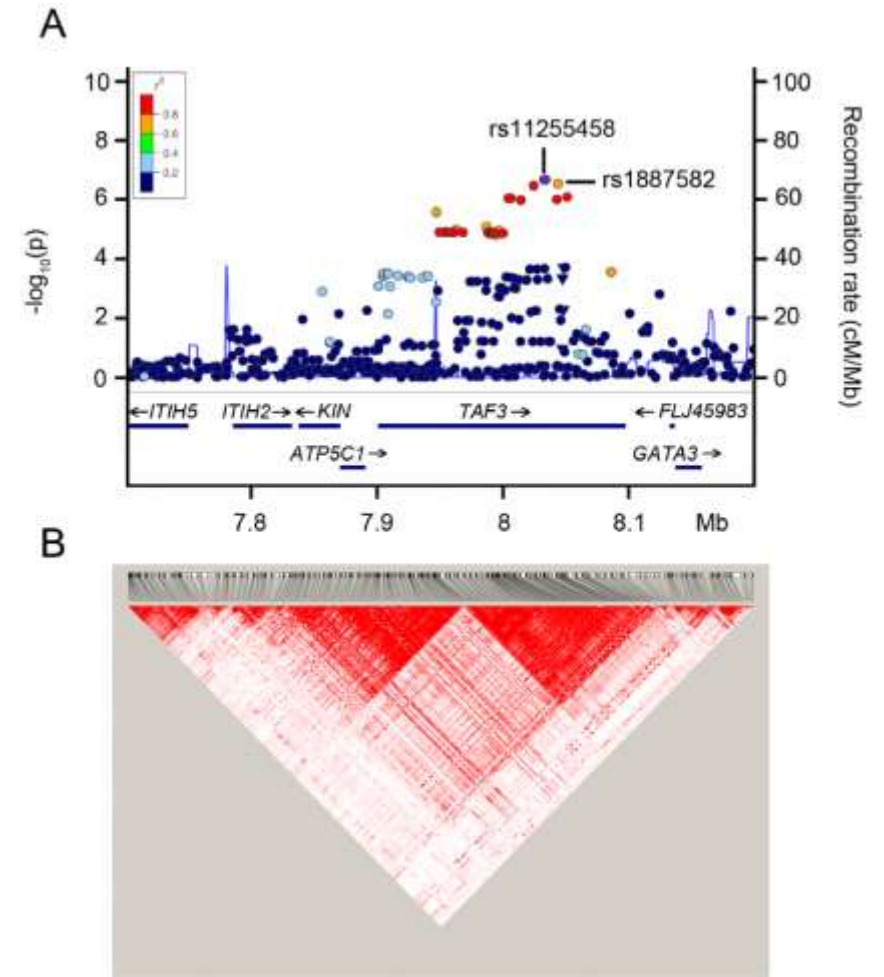
How to select variants influencing complex traits?

Selection of variants for PGS calculation

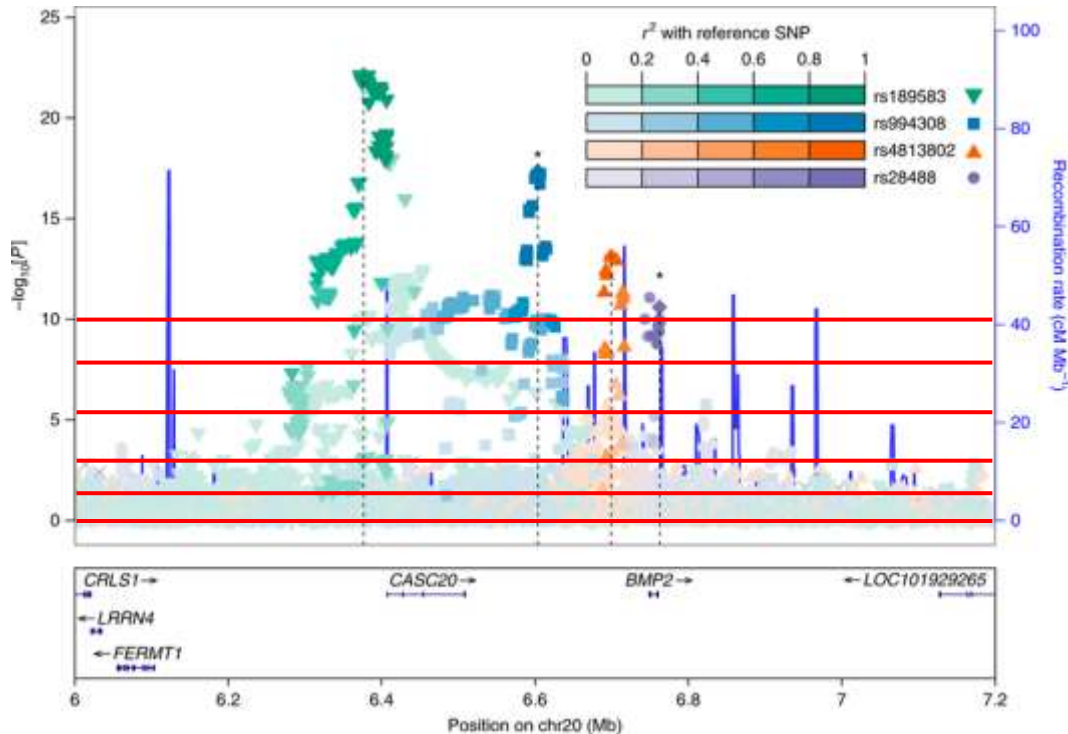
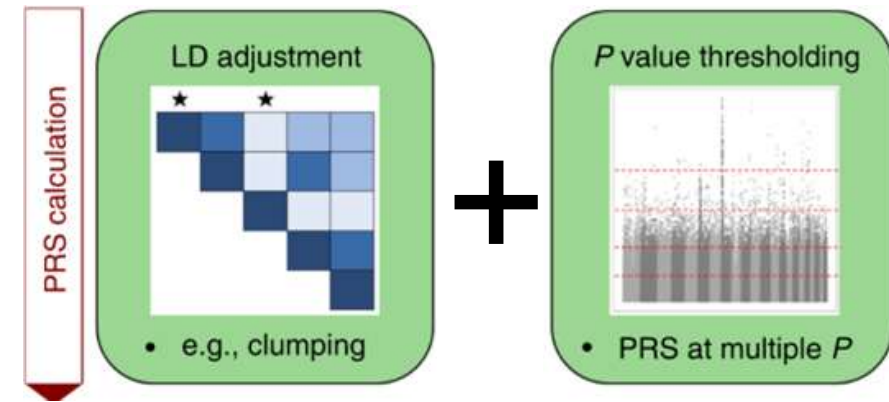
- Historically: independent top variants only
 - Challenging in omnigenic and polygenic models
 - Reduced predictive power
- Solution: use all variants (omnigenic model)
 - Problem: linkage disequilibrium (LD)

What is LD?

- Now: select **independent variants** (clumping, pruning)
 - One representative for each LD block
 - No overweighting of LD blocks



Clumping + Thresholding (C+T)



YES

P-value aware LD pruning

1. Select SNPs with a p-value < threshold
2. Select top associated variant
3. Remove all variants in LD with this SNP

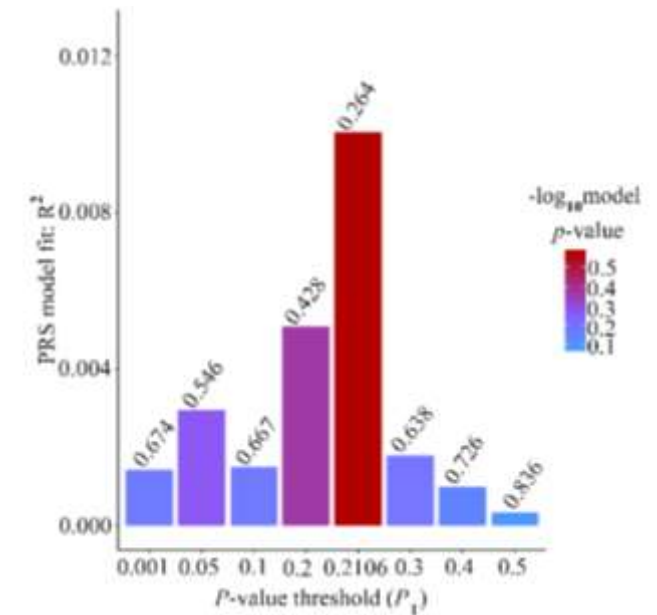
Any other significant variant left in the block ?

NO

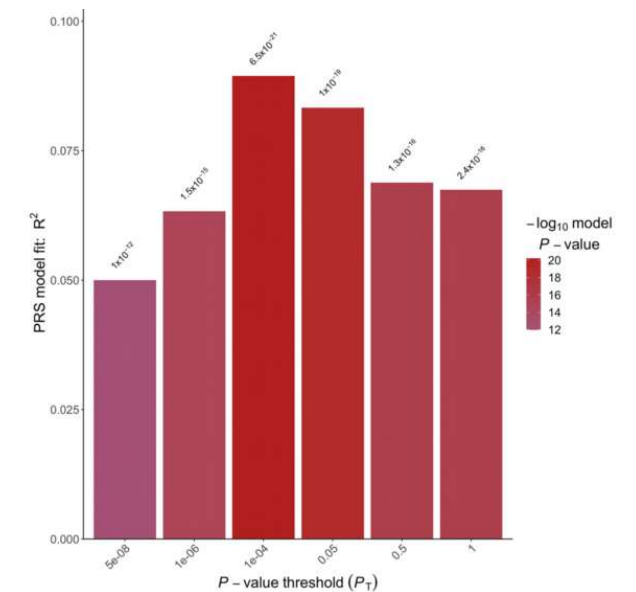
Set of pruned and significantly associated variants

Clumping + Thresholding (C+T)

- Which significance threshold to use to include variants?
 - Optimal threshold depends on the trait
 - More polygenicity → less stringent threshold
- Unknown beforehand
 - Try multiple values with validation
 - Integrated into PGS calculation software, e.g. *PRSice*



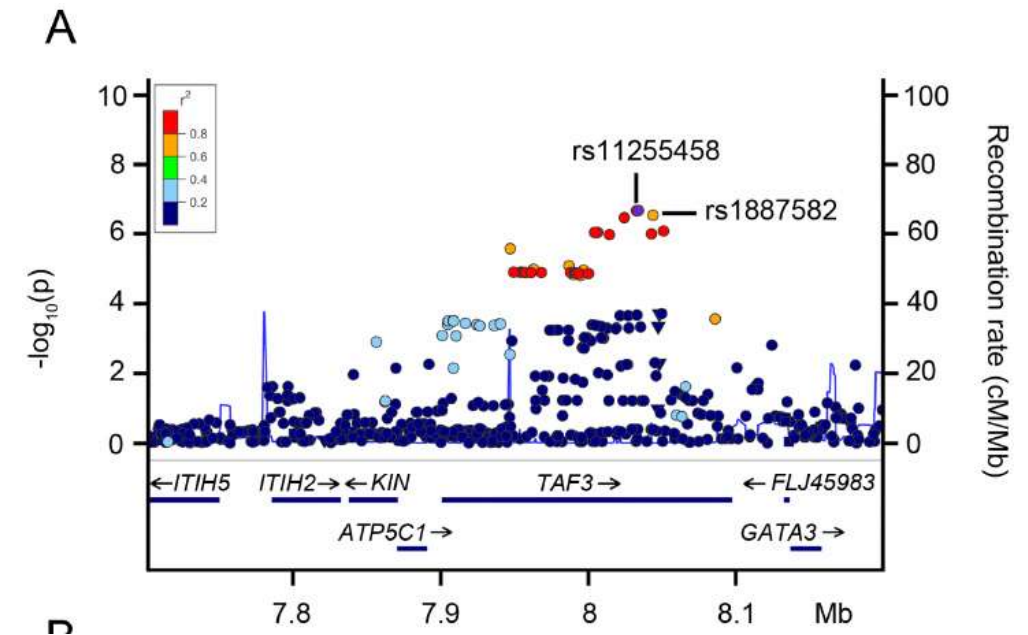
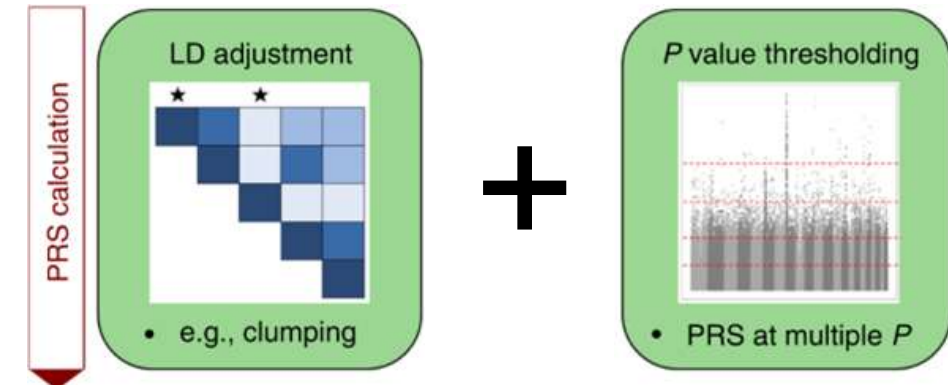
Wang et al. *Frontiers in Genetics*, July 2019



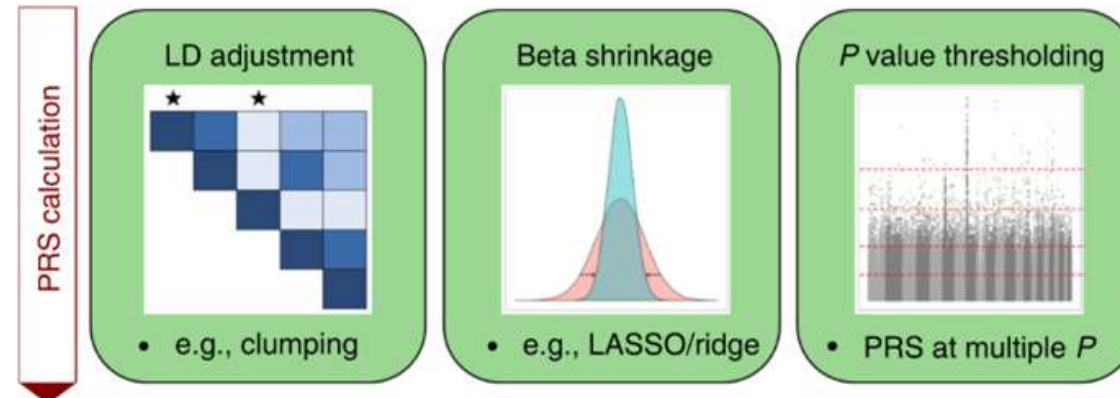
Maj et al. *Frontiers in Cardiovascular Medicine*, Feb 2022

Limitations of C+T

- Potential removal of secondary signals
- Based on the p-values but not the effect sizes
 - The p-value is related to the power of the study
 - Can miss low-effect variants in small sample sizes
- Ideal model = 'whole-genome' model
 - Account for LD
- Sample size is still a limiting factor for improved methods



Bayesian sparse regression methods (beta shrinkage)



- C+T: find subset of variants that best describe the trait of interest
- Beta shrinkage methods: find optimal transformation of the vector of effect sizes from GWAS to best represent the trait

$$PRS = \sum_{m=1}^M E\{\beta_m | Data\} G_m = \sum_{m=1}^M \widehat{\beta}_m$$

Bayesian sparse regression methods (beta shrinkage)

$$PRS = \sum_{m=1}^M E\{\beta_m | Data\} G_m = \sum_{m=1}^M \widehat{\beta}_m$$

- Models the distribution of shrunk/re-weighted effect sizes
- Uses:
 - prior that reflects the genetic architecture (e.g. all SNPs have non-zero weight)
 - genome-wide LD matrix to weigh down variants

→ Shrinkage method that produces scaled effect sizes genome-wide

- Downsides: too many hyperparameters → harder to interpret

Short list of software to calculate PGS

Clumping + thresholding

- PRSice

Bayesian sparse regression method

- LDpred: Vilhjalmsen, 2015
- SBayesR: Ge et al, 2019
- PRS-CS: Zeng et al, 2017

3

Polygenic scores

3.3

Application



Applying PGS



$$PGS_i = \sum_{j=1}^{N_{snps}} G_{ij} * \beta_j$$

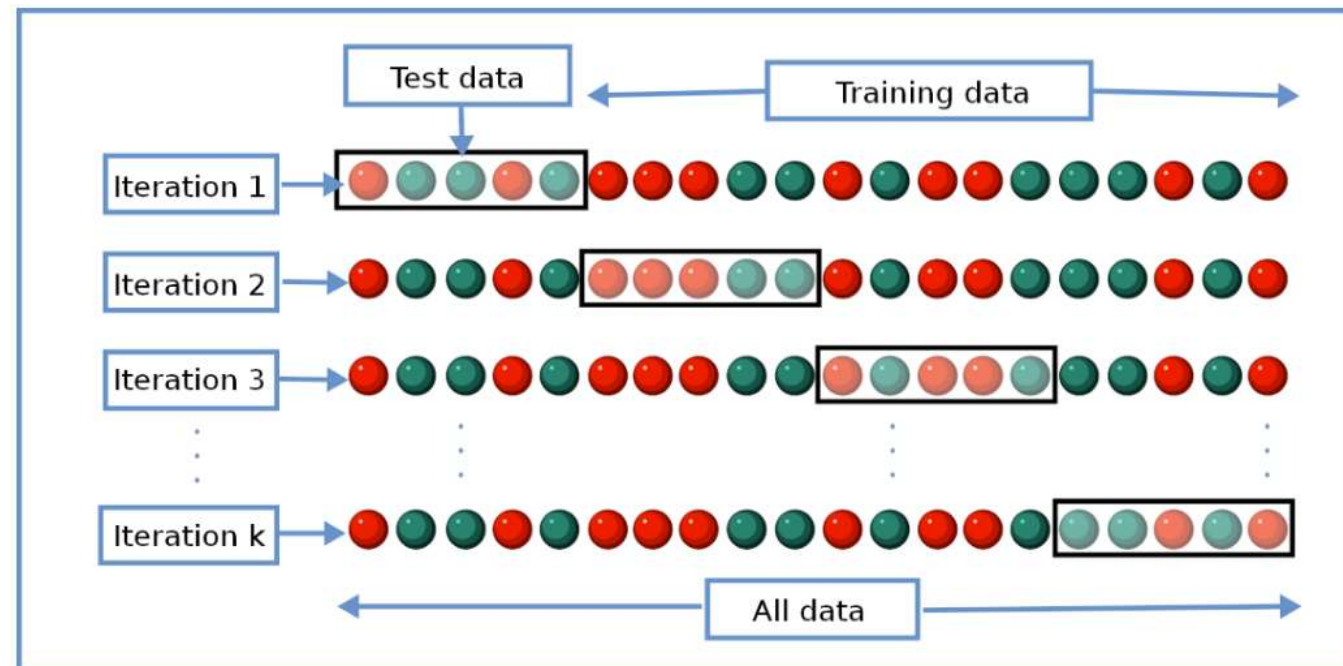
- Alleles must be matched between base and target data → beta inversion

$$\begin{aligned}\beta_{rs1234,A} &= 1.56 \\ alleles_{rs1234} &= \{A, T\} \\ \Rightarrow \beta_{rs1234,T} &= -1.56\end{aligned}$$

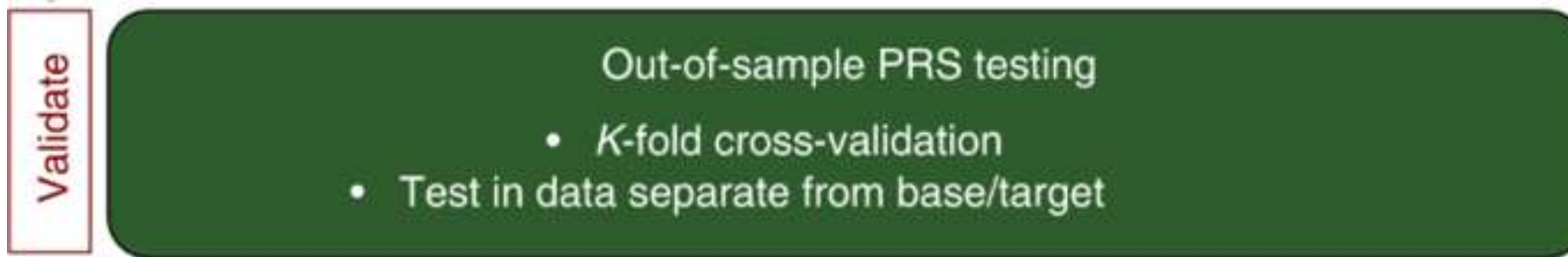
- K -fold cross-validation
- Test in data separate from base/target

Validation of PGS

- K -fold cross-validation
 - When no independent dataset available
 - Divide the sample in training and validation data
 - Repeat multiple times



Validation of PGS – independent sample



- Values to assess the prediction of PGS:

→ R²: amount of phenotypic variance explained by PGS (continuous traits)

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

Variability in dependent variable
not predicted by the model

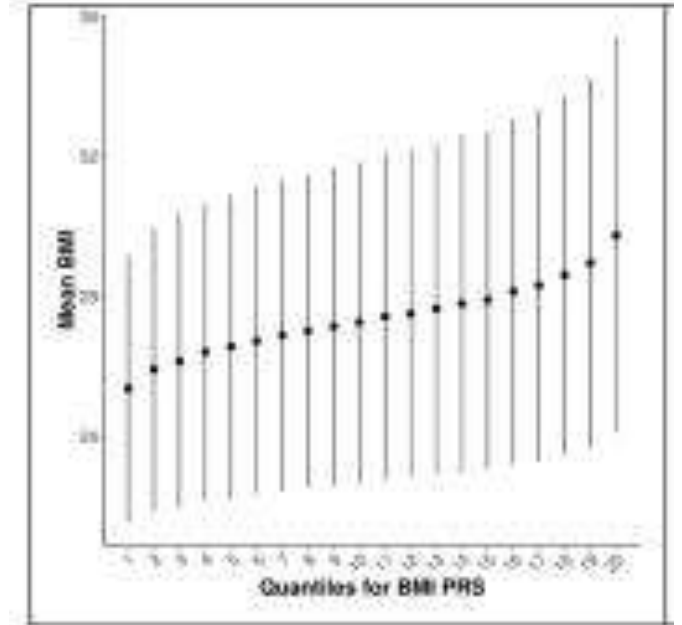
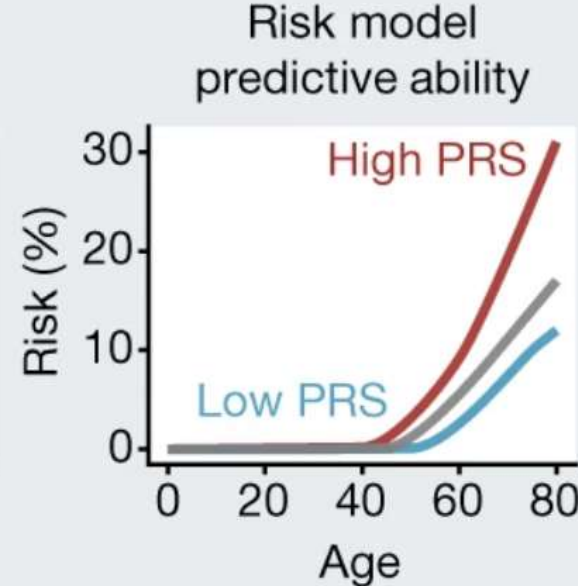
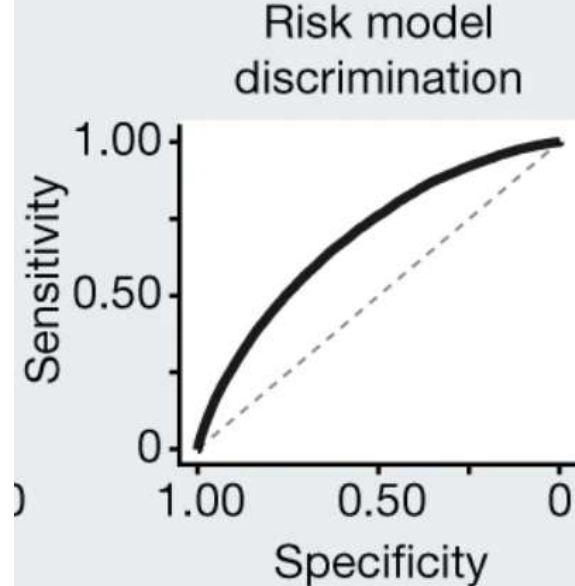
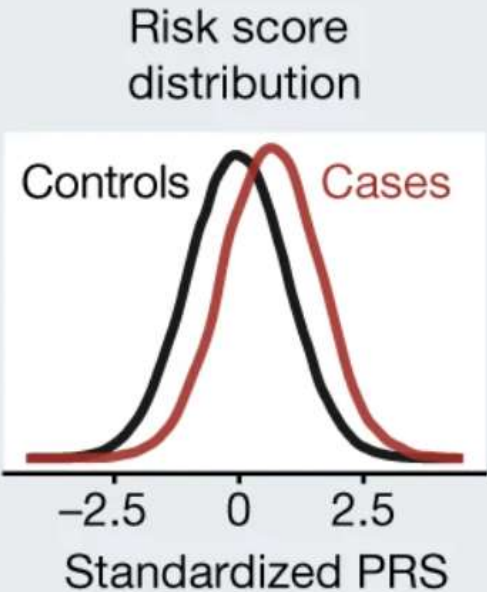
→ Pseudo-R²: R² for binary traits

→ Odds ratio between different groups

→ Area under the curve...

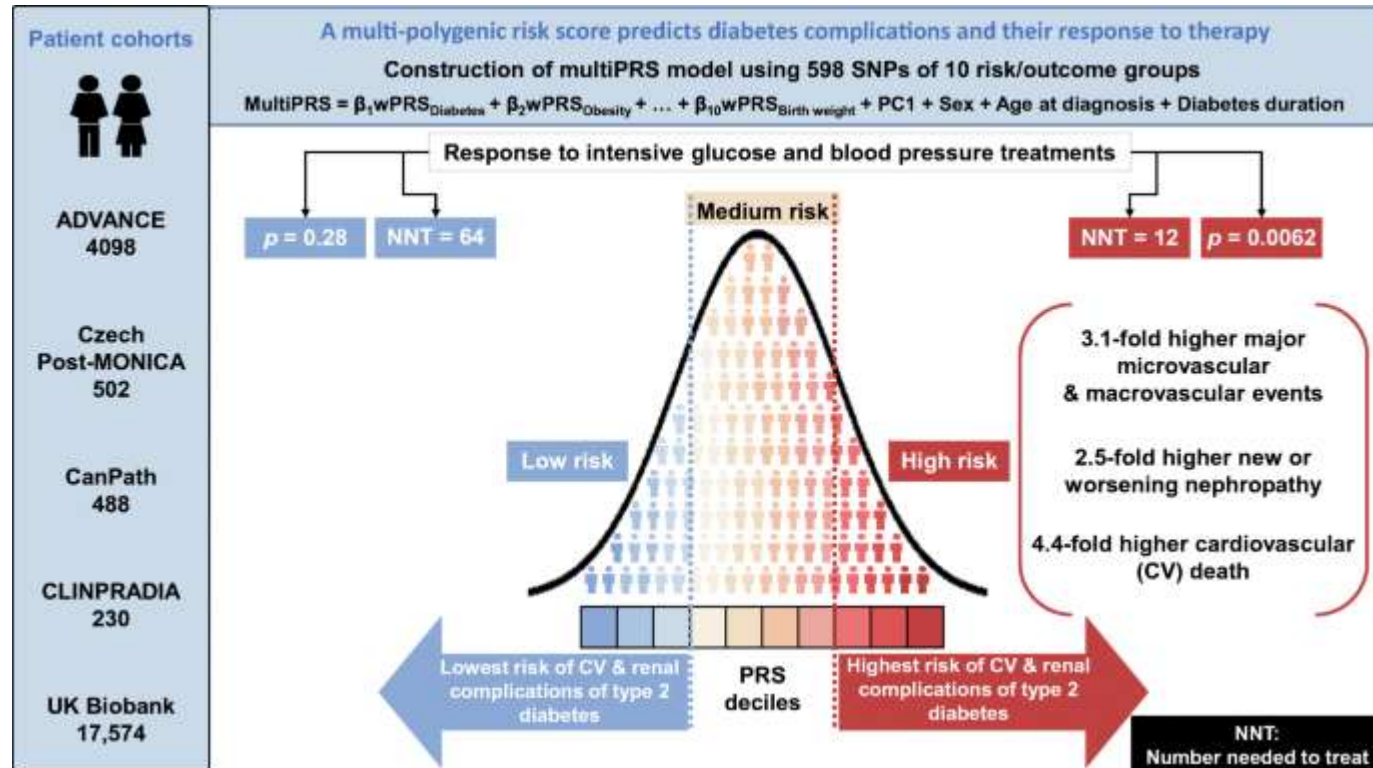
Variability in dependent variable

Validation of PGS - visualization



- **ROC curves:** Measure of discrimination in disease prediction
- **Incidence plots:** changes in OR in each quantile compared to the reference
- **Quantile plots:** changes in OR in each quantile compared to the reference

Clinical utility



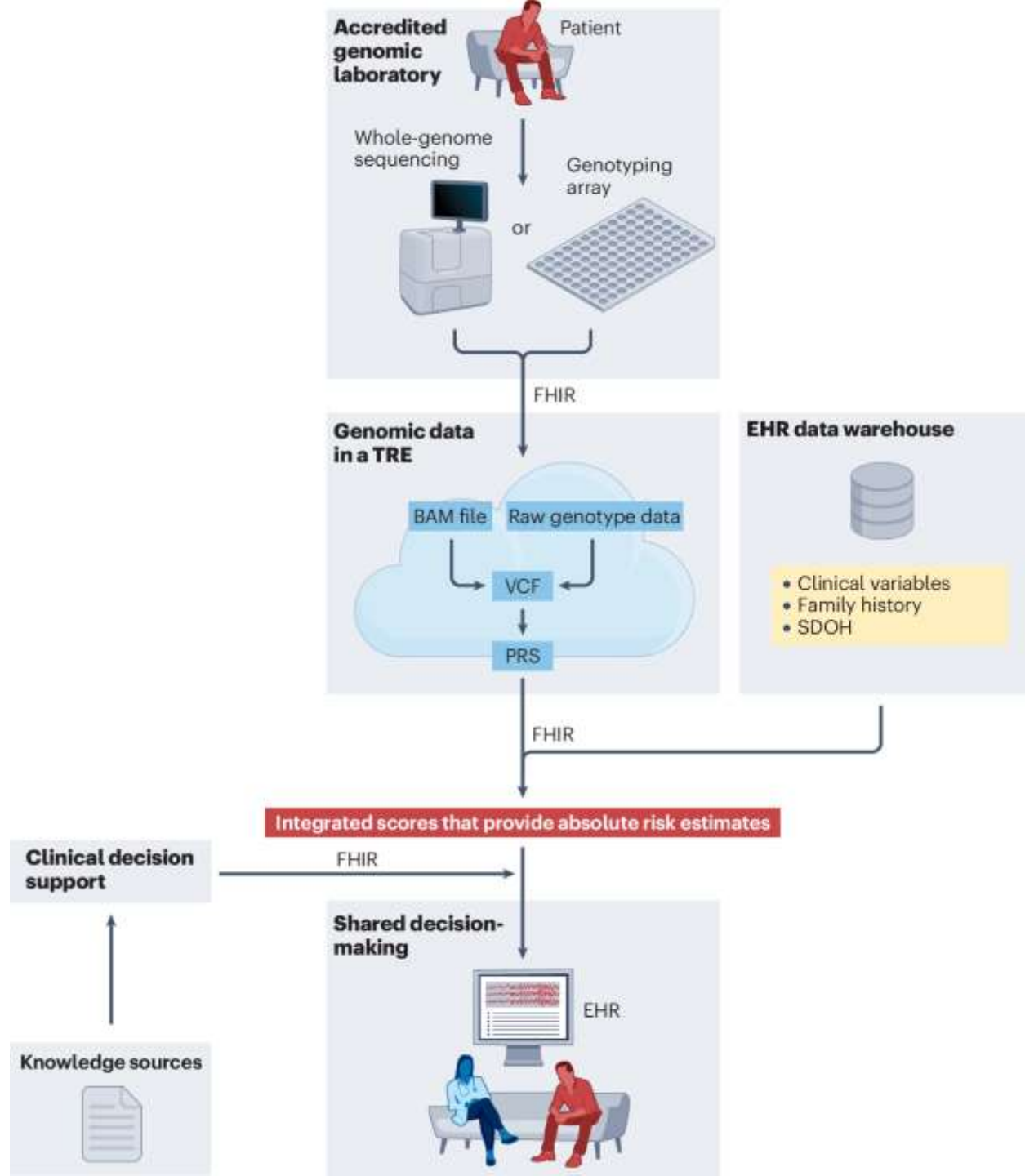
This novel multiPRS model stratified individuals with type 2 diabetes according to risk of complications and helped to target earlier those who would receive greater benefit from intensive therapy.

However!

- Even in their optimal form, PGS cannot be regarded as standalone predictors of disease onset.
- Instead, they are one component in a multifactorial framework for risk stratification that incorporates clinical variables, family history, lifestyle and environmental exposures and other omics data.
- The complex aetiology of common diseases requires integrative risk models that reflect both genetic predisposition and environmental context.

Kullo, I.J. Clinical use of polygenic risk scores: current status, barriers and future directions. *Nat Rev Genet* (2025). <https://doi.org/10.1038/s41576-025-00900-8>

Clinical utility



3

Polygenic scores

3.4

Limitations



Limitations of PGS

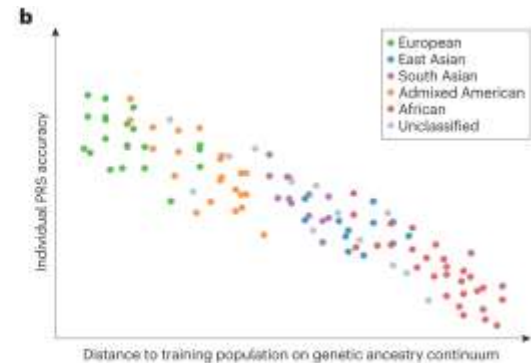
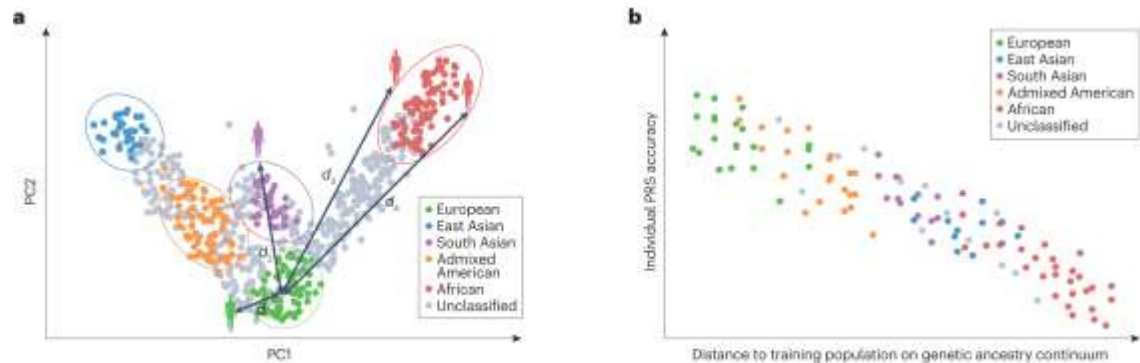
No environmental factors considered

Dependent on homogeneity in discovery and testing samples

Predictive power depends on sample size

- Low predictive power → limited clinical use
- Low transferability when deviation from original GWAS cohort (e.g. ancestry)

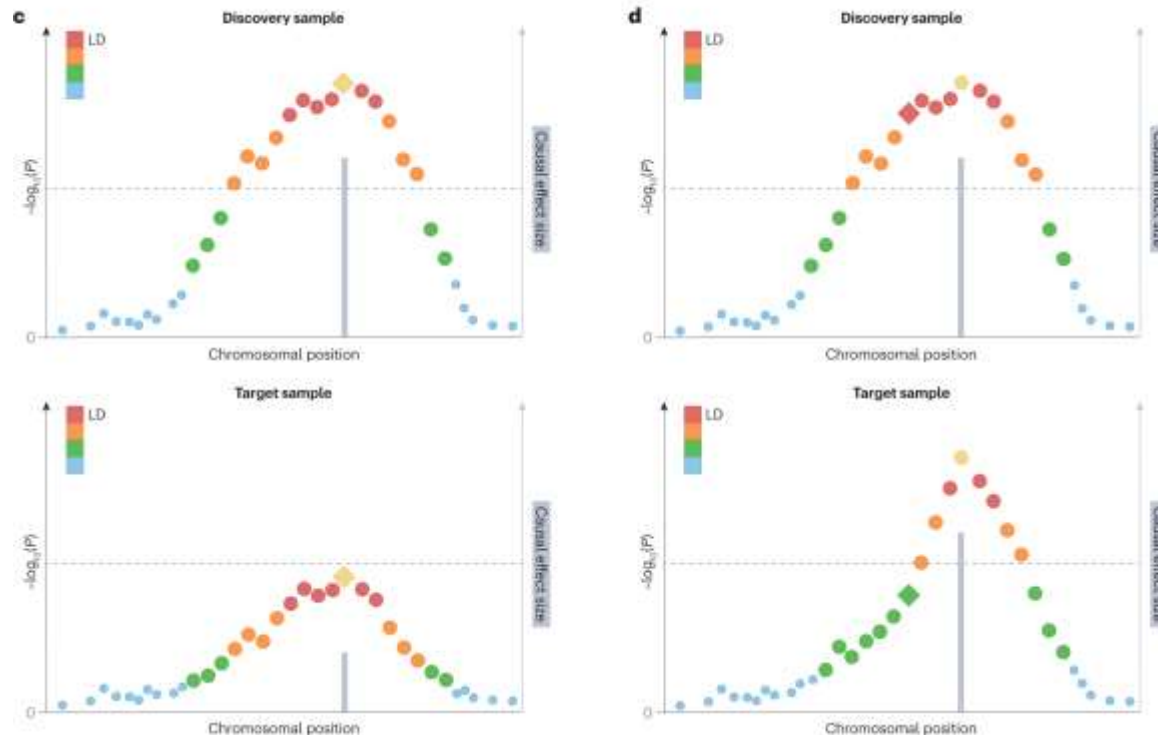
Genetic factors that can influence PGS performance.



The issue: the accuracy of genomic prediction decreases as the genetic distance between the training and target populations increases (a & b).

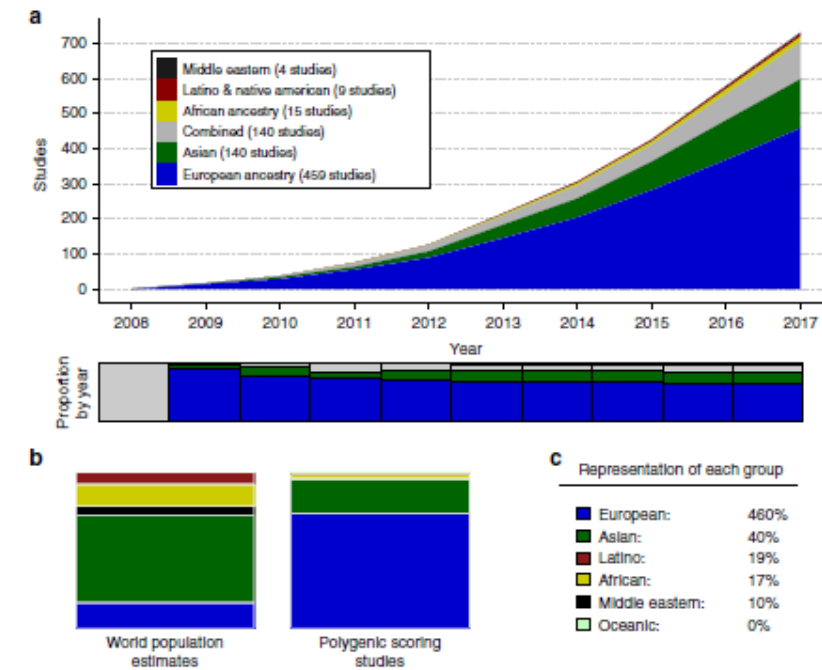
Genetic factors influencing PGS transferability across populations:

- differences in heritability
- differences in causal allele(s)
 - allelic frequencies
 - allelic effect sizes
- linkage disequilibrium (LD) patterns

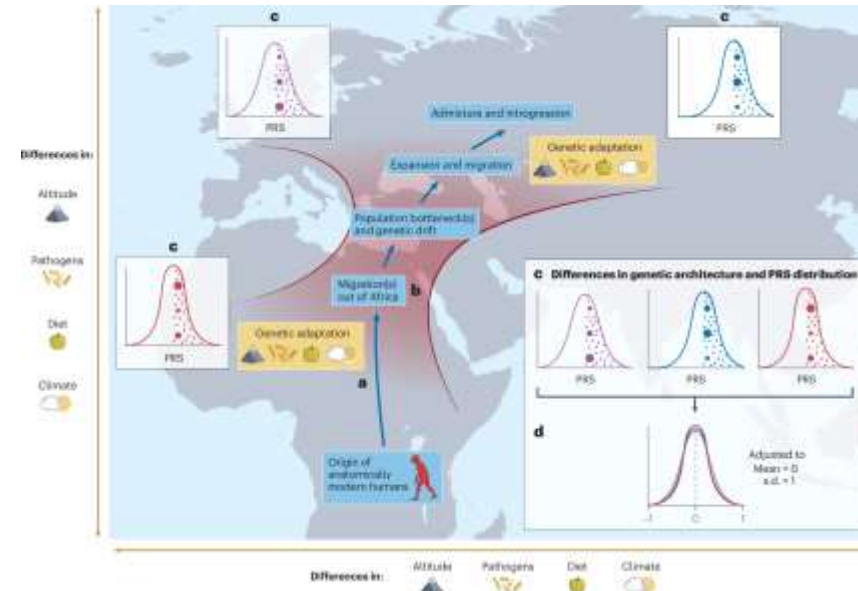


Trans-ancestry PGS

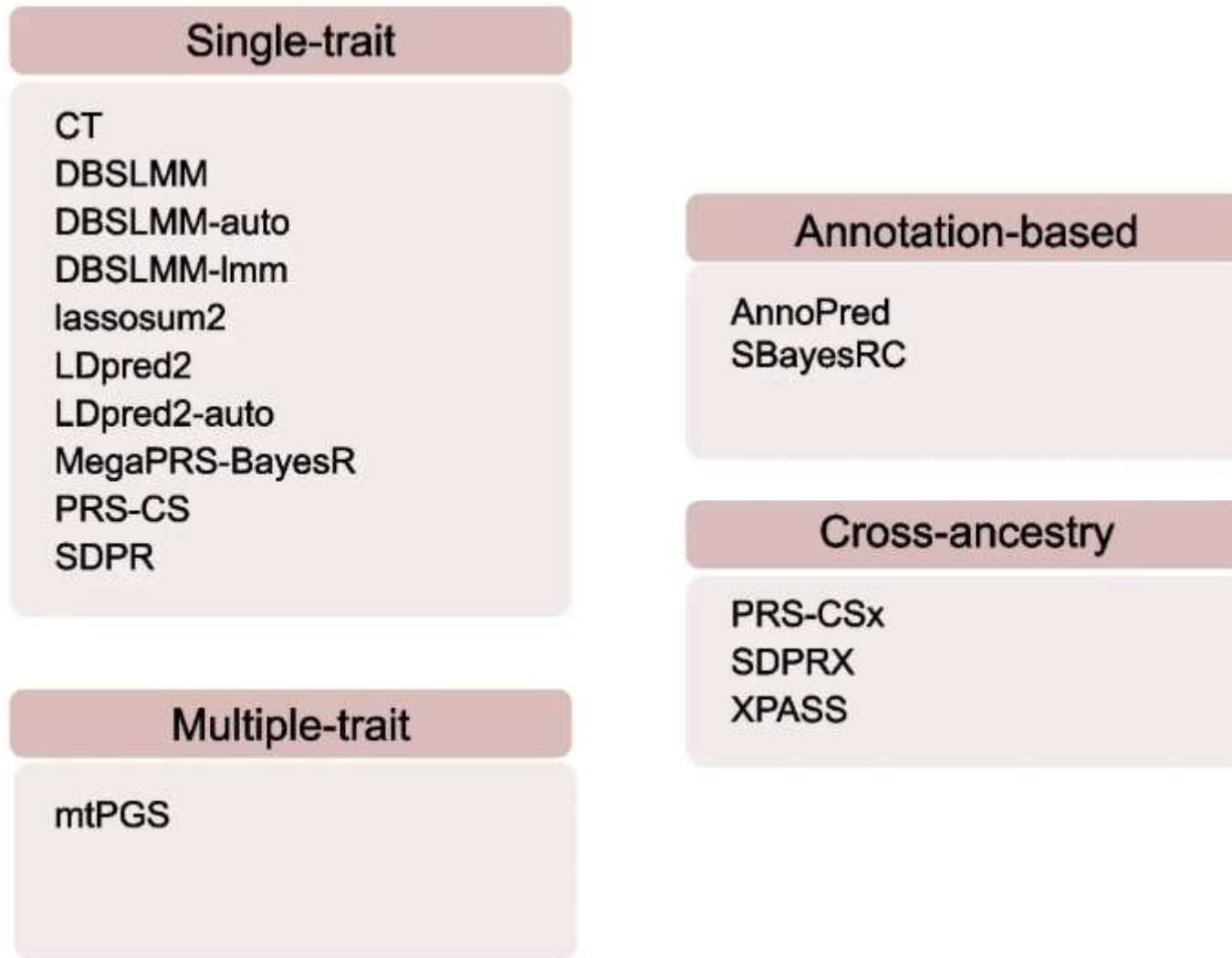
- Currently, PGS mainly derived from European populations
- Poor transferability to non-European populations due to differences in:
 - Allele frequencies
 - LD
 - Effect sizes
 - Environmental factors
- Non-European PGS are limited due to small sample sizes
- Trans-ancestry PGS = active area of research
 - meta-regression, ...
 - Decrease health disparities



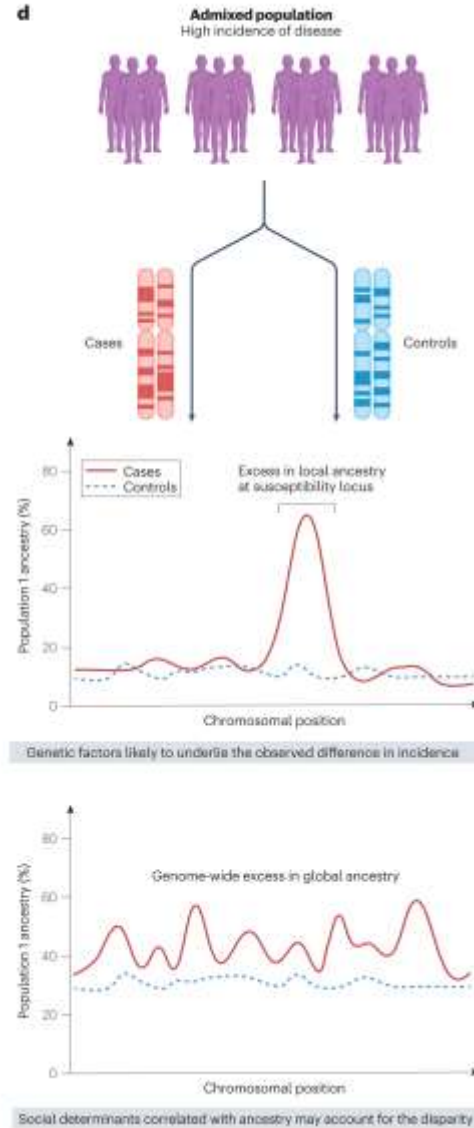
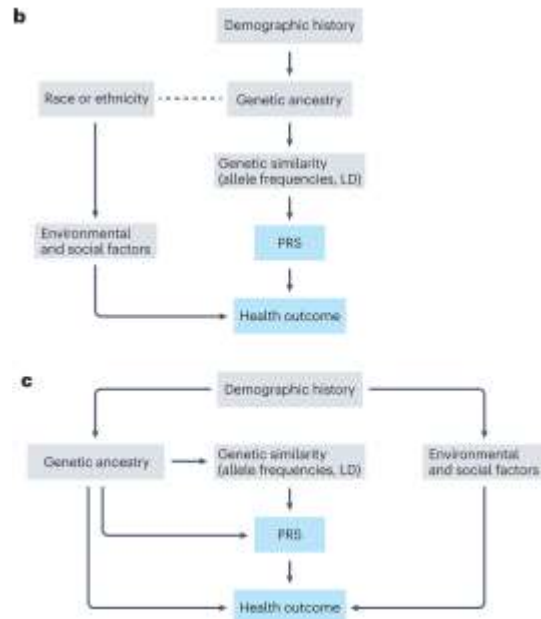
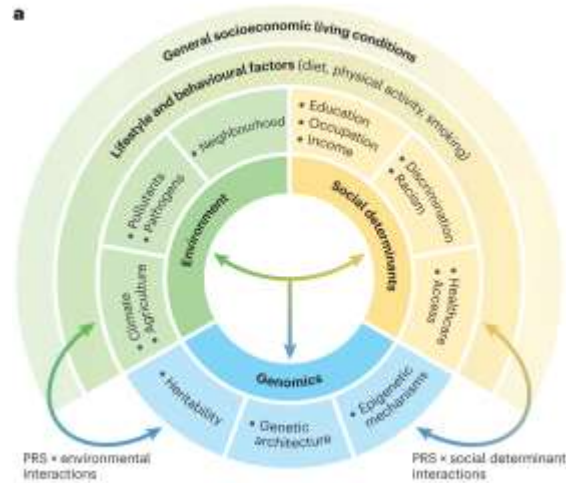
Duncan et al. Nat. Comm. 2019



Methods to improve PRS transferability across populations.



Interplay between social, environmental and genetic determinants of health.

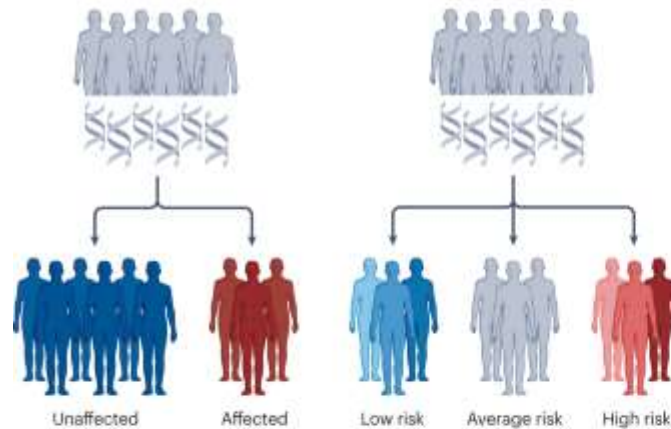


Residual confounding may bias PRS associations when genetic ancestry is correlated with environmental and/or social factors due to shared demographic histories.

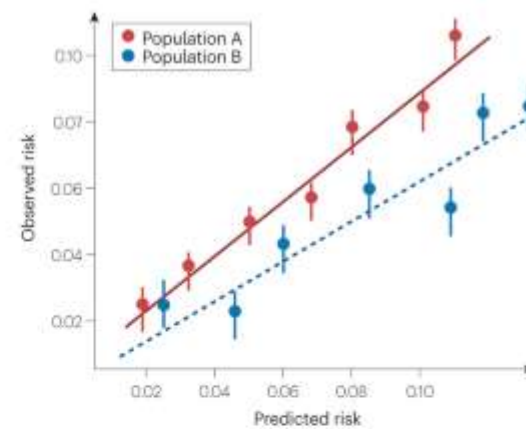
For instance, when asthma is the health outcome and exposure to air pollution is the non-genetic risk factor, standard methods such as PCA may under-correct for population structure.

Considerations for the assessment of PRS clinical utility.

a Risk prediction versus risk stratification

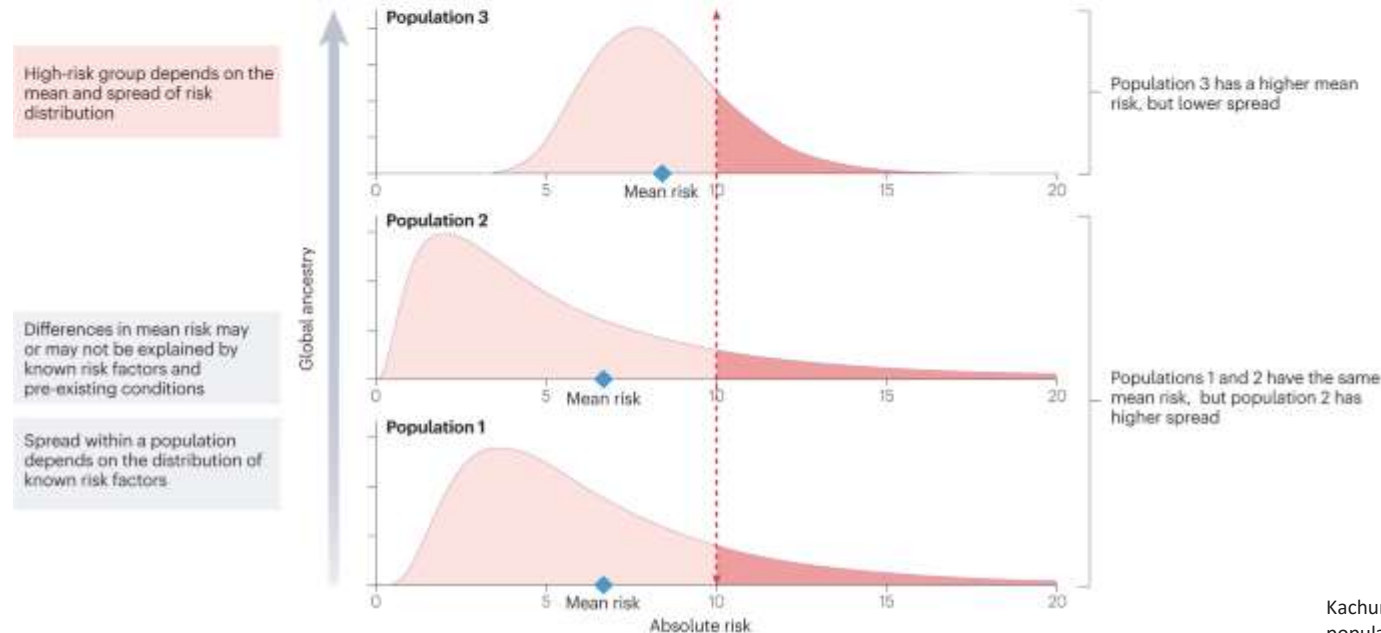


b Model calibration



- a) Visual representation of the difference between risk prediction and risk stratification.
- b) An example of model calibration, that is, the agreement between observed and estimated disease risk. Accurate estimation of absolute risks requires well-calibrated models. For instance, risks are systematically overestimated for Population B compared with Population A.
- c) Cross-population calibration of the polygenic risk score (PRS) distributions, which can have different mean and spread. Cross-population calibration is important when selecting a single cut-off to identify individuals as high risk across samples with diverse ancestral and sociocultural backgrounds.

c Cross-population calibration



HELMHOLTZ MUNICH

Conclusions

- 1) Substantial efforts are needed to improve PRS transferability, integrate PRSs into routine health care and equitably deliver PRS to global populations.
 - Novel statistical and computational methods are needed to construct, validate and optimize PRSs that can be applied to any individual along the continuum of genetic ancestry.
 - Admixture-aware and clinically informative metrics are also needed to assess the accuracy, calibration, uncertainty and stability of PRS prediction in ancestrally diverse samples.
 - Develop integrative clinical models that combine PRSs with environmental established risk factors for patients of diverse ancestral and sociocultural backgrounds.
- 2) Encouragingly, the field is rapidly advancing on all fronts, including method development, data generation and clinical implementation. For example, the National Institutes of Health (NIH)-funded Polygenic Risk Methods in Diverse Populations (PRIMED) Consortium (<https://primedconsortium.org>) is developing methods and pooling genomic and phenotypic information to improve the use of PRSs for disease prediction in diverse populations.
- 3) When combined with other modalities — such as proteomics, metabolomics, imaging and longitudinal EHR data — PRSs have the potential to markedly improve disease risk prediction, inform screening strategies and support earlier interventions to reduce risk.

3

Polygenic scores

3.5

Workshop



Timeline

- Introduction (Exercise 1): 10 minutes
- Manual score in R: 30 minutes (Exercises 2-5)
- Score in Plink: 20 minutes (Exercises 6-7)
- PGS and Polygenicity: 20 minutes (Exercises 8-9)



Thank you.